门动作磁耦合压电采能节点
详细数字工程与验证报告

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结论边界：本案例尚未通过一次开门冷启动要求；未制造、未烧录、未进行实物测量。

本报告关联同一个 D2 设计版本的参数化 CAD、门体动画、COMSOL 原生模型与场图、机电降阶计算、厂商电源模型、PCB 铜线、目标固件和验收记录。文中展示实际计算结果，并保留未通过的项目；模型和图像不是实物性能的替代证明。



图 1: 900×2100×40 mm 门的安装场景。产品在门框上，驱动磁体随门运动。

设计任务	从静止、储能电容零电压开始，一次开门产生可接收无线事件。
当前结果	数字工程已建立；当前 D2 供能不足，未达到该任务要求。
通过范围	CAD 实体、采样运动间距、降阶时间步、负对照、独立 PCB 检查、目标固件重复编译。
未闭合范围	冷启动、磁场网格、原生 EDA 验收、真实硬件及双向电源耦合。

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1 需求、参考来源与版本选择

该产品是安装在门框上的压电采能无线动作节点。驱动磁体安装在门上，随开门运动改变梁端磁体所受磁力；压电梁将机械响应转为电能，经 LTC3588-1 整流、储能和稳压后，供 nRF52832 模块发送 BLE 事件。

本轮以三篇 ViPSN/拨动压电跨域仿真论文作为参考材料，以厂商资料确定电源和无线模块的接口。论文中的性能没有直接复制为本产品性能，缺少实测的阻尼、粘接、磁材离散性均作为假设记录。没有训练神经网络：当前功能是二值动作事件，没有支持方向分类或 NAS 的数据基础。

参数	本轮固定值或验收标准
正常输入	门角 0–65°，2 s 开门，继续观察 2 s
弱输入	同样角度，8 s 慢开门
初始条件	电源电容零电压；机械闭门磁预载单独计入初始能量
任务验收	一次开门达到冷启动并支持至少一次无线事件；未通过
数值验收	磁力网格差异不高于 5%；时间步能量差异不高于 5%
降阶范围	梁端位移绝对值不超过 8 mm；仍需后续几何非线性校核
电路验收	独立几何检查和原生 EDA ERC/DRC 分别记录，不相互替代

D1 驱动磁体距离过近，受力约 4 N，超出当前梁模型允许范围。D2 将磁体中心横向间距增至 48 mm，以使运动计算处于声明的范围内。这个改动降低了采能能力；不能把“避免模型越界”写成“产品任务通过”。冷启动标准没有降低。

2 机械尺寸、坐标与装配

门沿世界坐标 z 轴绕铰链旋转，铰链位于 $x = 0$。产品局部坐标以 mm 为单位，安装转换为

$$(x_w, y_w, z_w) = (1004 - x,\ 70 + z,\ 2140 + y).$$

固定产品与运动驱动磁体采用同一个变换，Blender 没有另造不同尺寸的展示物。

零部件	D2 几何定义
门	900×2100×40 mm
主壳体	PETG，主体 136×62×66 mm，名义壁厚 2 mm
含安装耳壳体包络	STEP 回读约 136.003×73.002×66.003 mm
全机构包络	含驱动件约 188.002×82.000×75.001 mm
盖板	PC，厚 2 mm
铜基梁	100×10×0.53 mm；夹持 4 mm，自由长 96 mm
压电片	PZT-5A 名义材料，50×8×0.30 mm，梁上 $x = 10$–60 mm
端部磁体	10×10×10 mm，中心 $(100, 0, 5.265)$ mm
驱动磁体	20×20×20 mm，中心 $(148, 0, -25)$ mm
PCB	80×30×1.6 mm，中心 $(45, -12, -22)$ mm
PCB 安装孔	直径 2.4 mm，板坐标 (4,4)、(76,4)、(4,26)、(76,26) mm

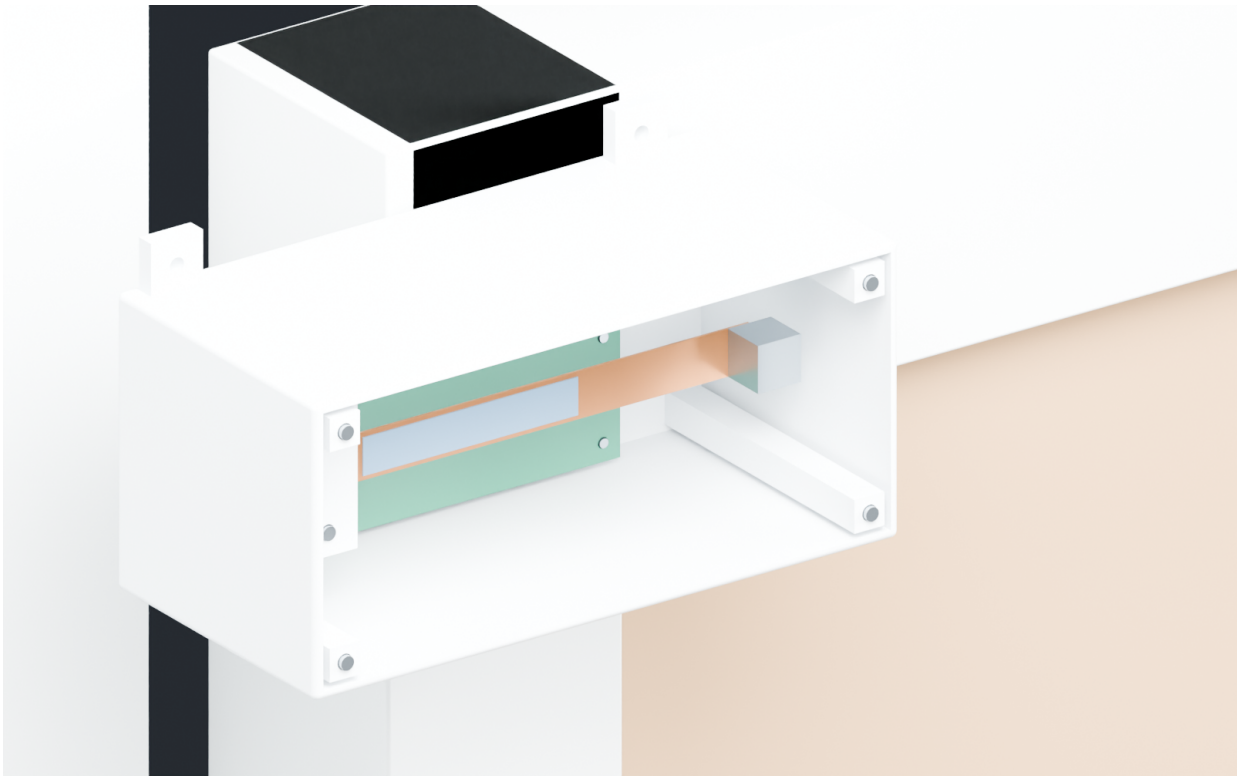


图 2: 由实际 CAD 零件导入的机构局部视图；PCB 在壳体内，门上驱动件与固定梁组件分开。

3 CAD 图纸与安装检查

本轮输出 22 个产品实体的 STEP、BREP、STL、零件明细、装配 STEP 和安装 STEP。实体回读有效，静态无未授权实体干涉。附录包含 22 张由原生几何投影生成的 A3 零件图，每张包括三视图、等轴测视图与包络尺寸。

这些图纸达到数字样机复核范围；尚未完成生产工程图的全部孔位基准、公差配合、粗糙度、螺纹、粘接厚度与材料采购等级标注。螺钉使用名义螺纹包络，不能据此声称完成量产制图。CAD 源参数和生成程序一起交付，可继续编辑。

运动检查使用真实 BREP，在 0–65° 每隔 1° 比较 4 个运动部件与 18 个固定部件，发现碰撞数为 0，最小间距为 20.6564 mm，限制对为 1° 时的驱动竖支架与壳体。该结果覆盖离散刚体样本，不覆盖连续扫掠、柔性梁、导线、整个建筑门框和装配公差。

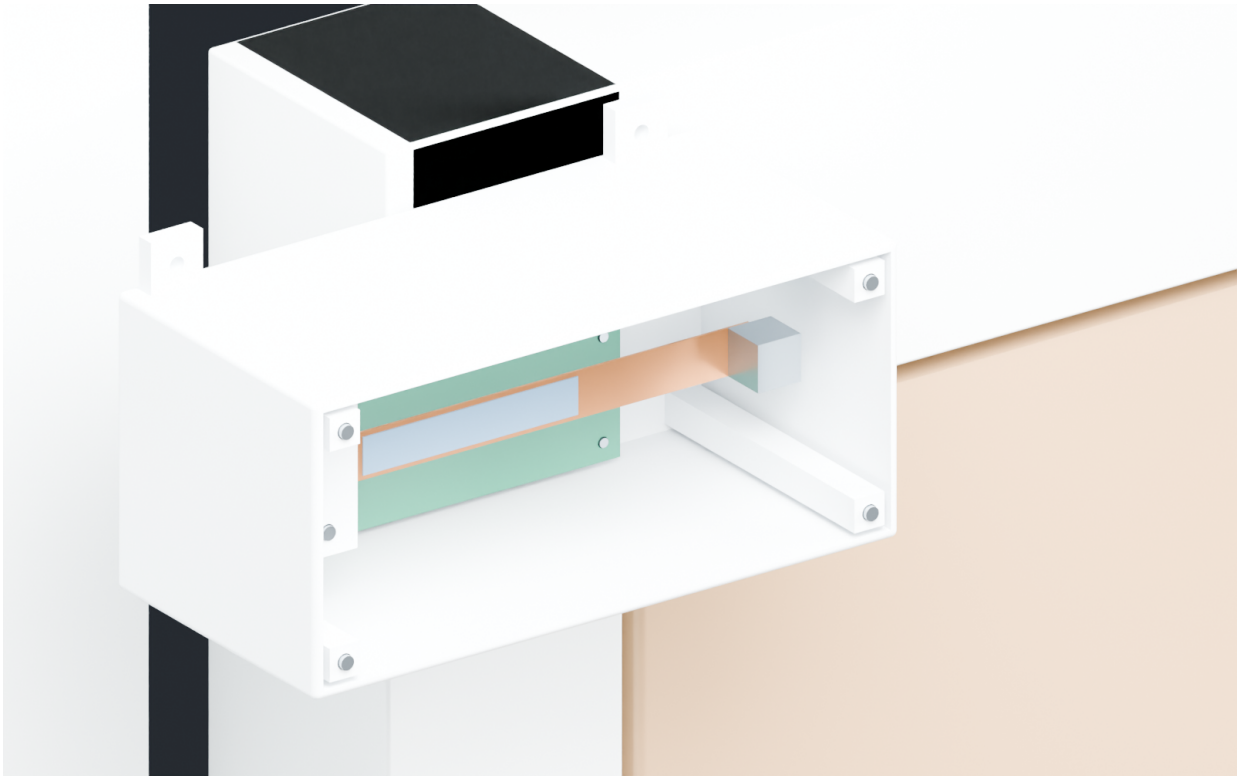


图 3: 打开后的局部机构视图。动画反映刚体安装和门输入；梁振动没有伪造为仿真动画。

4 COMSOL 压电梁模型

原生 COMSOL 6.2 模型耦合 Solid Mechanics、Electrostatics 与 Electrical Circuit。铜基梁使用 $E = 110$ GPa、$\rho = 8960$ kg/m^3；压电层使用模型库 PZT-5A，极化沿 $+z$。当前采用理想粘接与线性材料，未测量胶层、压电常数离散与疲劳寿命。

静力载荷为 0.3 N，用来提取刚度及进行场分布检查，不能当成完整开门过程峰值应力。

网格级别	最大位移/mm	梁端位移/mm	最大应力/MPa
6	3.72932	3.40423	62.0536
4	3.76237	3.43469	65.5216
3	3.76838	3.44031	67.5038

最后两级梁端位移差约 0.164%，应力差约 2.94%。这说明该静载下两级结果接近，不能保证夹持奇异点的点应力已达到材料强度评估要求。第一弯曲频率为 15.91524 Hz，后续模态包括 137.1092、142.1622、375.9652 Hz。

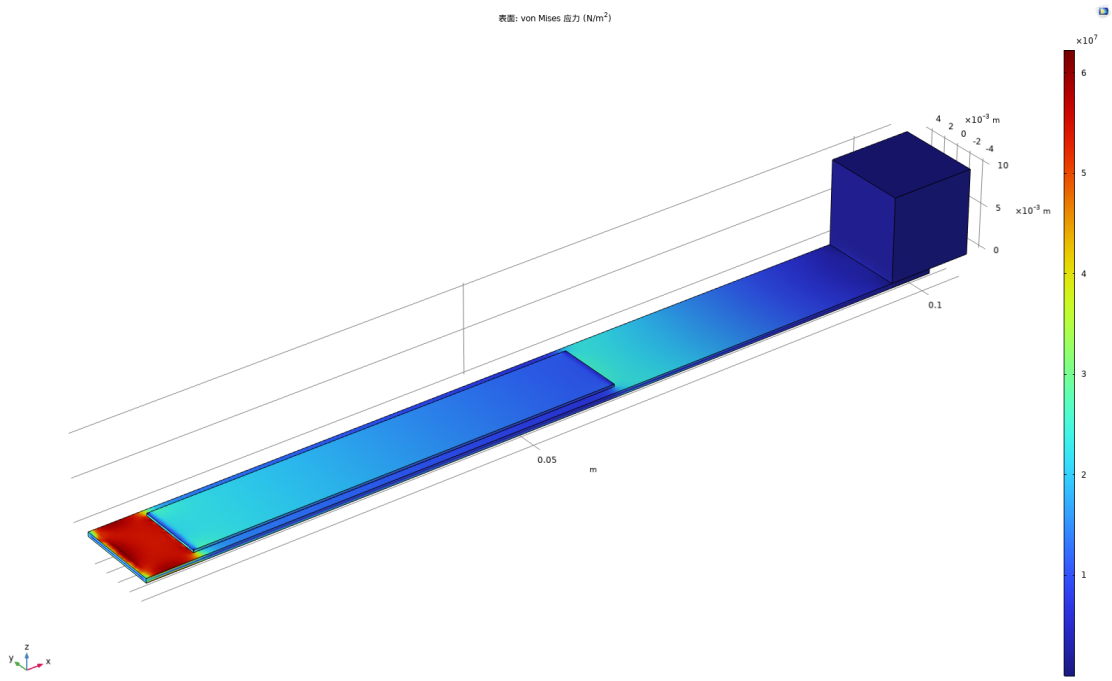


图 4: 原生 COMSOL 静载应力场，颜色由求解变量生成，并保留单位与色标。

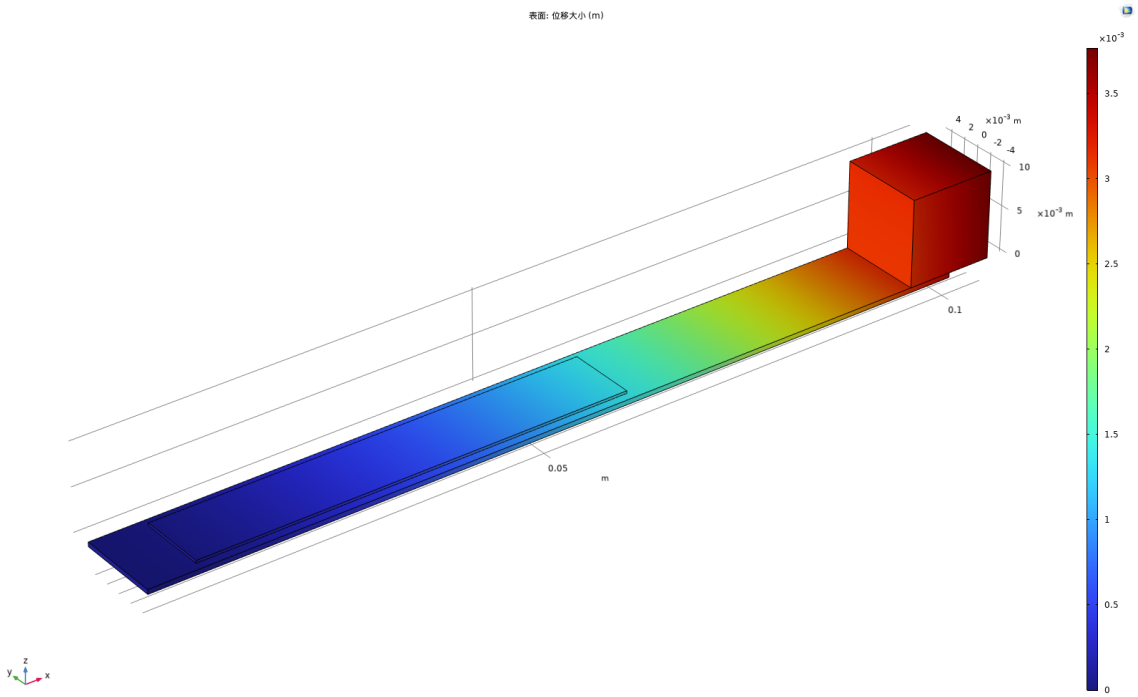


图 5: 同一静载模型的位移场。形变显示和颜色值的含义以原生模型设置为准。

降阶参数	由原生模型提取的数值
等效刚度 k	87.201538 N/m
等效模态质量 m	0.008720430 kg
机电耦合 θ	0.000216801 N/V
阻挡电容 C_p	17.23455 nF
阻尼比 ζ	0.02，工程假设；另计算 0.05 对照

5 磁场模型与网格限制

COMSOL Magnetic Fields, No Currents 用于求解驱动磁体场。驱动磁体按 N42 名义 $B_r = 1.25$ T、$\mu_r = 1.05$，空气域半径 150 mm。梁端磁体通过体积分 $M_x \partial B_x/\partial x$ 等梯度项估算受力。该模型没有自洽求解端部磁体对源场的反作用，也没有完整磁滞与退磁模型。

级别	z 偏移/mm	F_x/N	F_z/N
3	0	0.77928	0.03091
2	0	0.83613	0.00144
3	20	0.36158	0.39821
2	20	0.36559	0.39103
3	45	-0.02862	0.13680
2	45	-0.03273	0.13866

最大向量差异约 7.65%，超过 5% 门槛，因此磁力仍是未收敛工程估计。更细网格的未完成尝试保留在执行记录中，不能选取局部看起来较好的分量宣称通过。

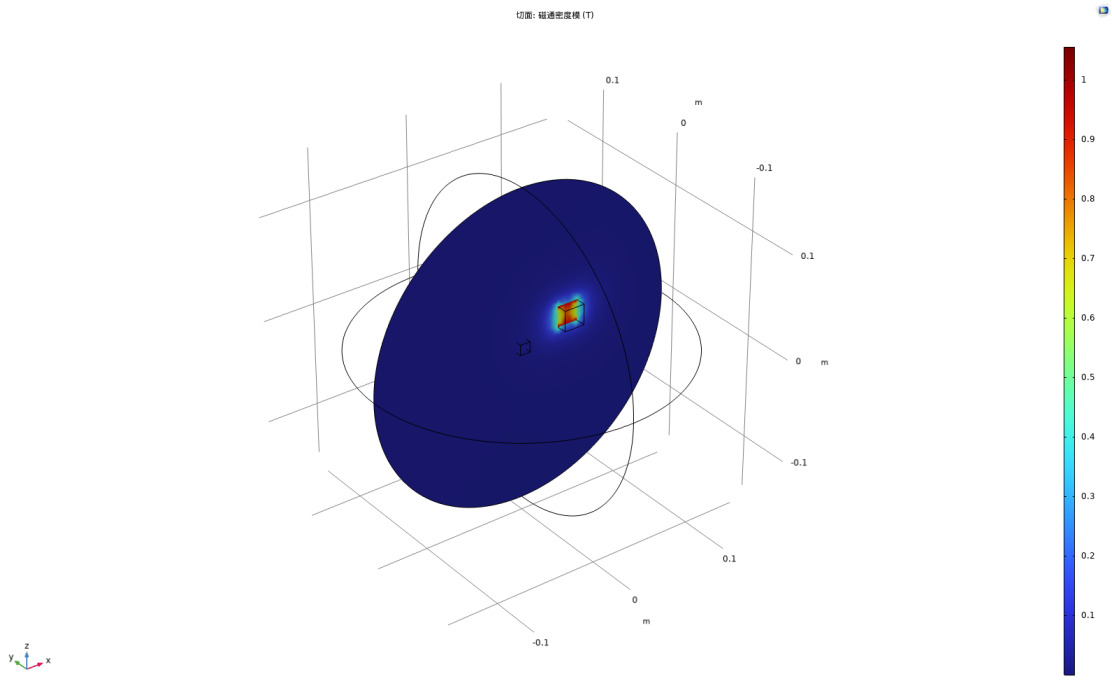


图 6: 原生三维磁场及切面场图。云图用于解释场分布，验收仍以数值和网格检查为准。

6 机电降阶与时间域结果

$$m\ddot{q} + c\dot{q} + kq + \theta V = F_{\mathrm{mag}}(\alpha(t), q), \qquad C_p\dot{V} + V/R = \theta\dot{q}.$$

有限体积磁荷积分构造力表，按门角和梁端位移插值；该计算采用均匀磁化近似，与 COMSOL 的 $\mu_r = 1.05$ 模型不是完全相同的方程。闭门初始预载约 -4.17178 mm，其弹性势能约 758.82 μJ 被显式记入能量账本，不能把这部分能量漏掉后虚增采能效率。

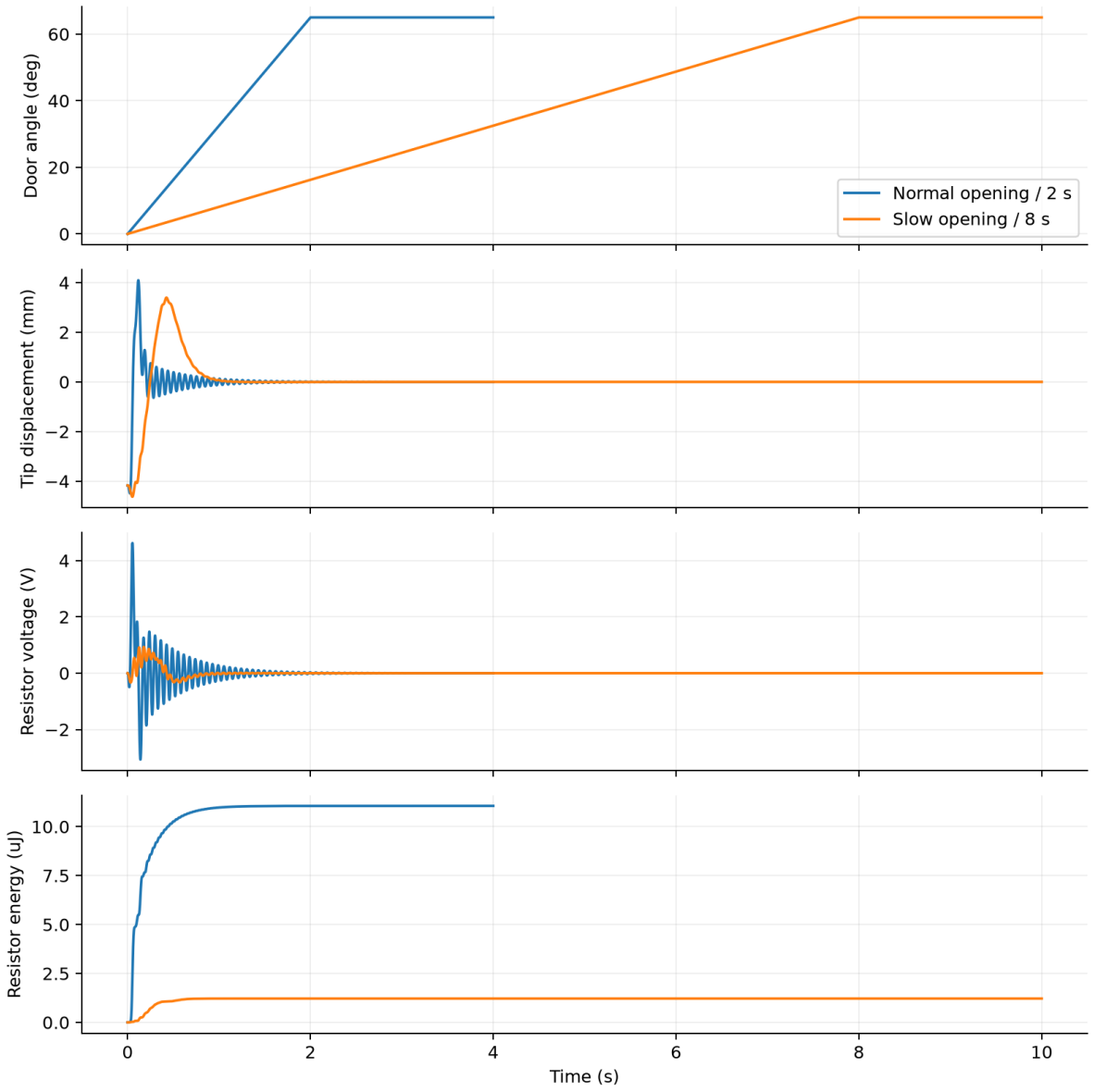


图 7: 实际输出 CSV 绘制的门角、梁端位移、电压及电阻能量。慢动作的电能更少。

工况	峰值位移/mm	峰值电压/V	电阻能量/μJ
正常，100 kΩ	4.49242	4.61972	11.06091
正常，1 MΩ	4.48946	31.83670	50.55926
慢开，100 kΩ	4.62830	0.93182	1.21786
高阻尼，100 kΩ	4.47914	4.42765	7.40968
无动作 / 无磁体	–	0	0

正常工况 1 ms 与 0.5 ms 的电阻能量相差约 2.17×10^{-6} μJ。0.5 ms 的最大能量残差为 5.60×10^{-8} J，相对参考能量约 7.38×10^{-5}。这些属于当前降阶模型的内部数值检查，不是实物校准。

7 电路原理与器件选择

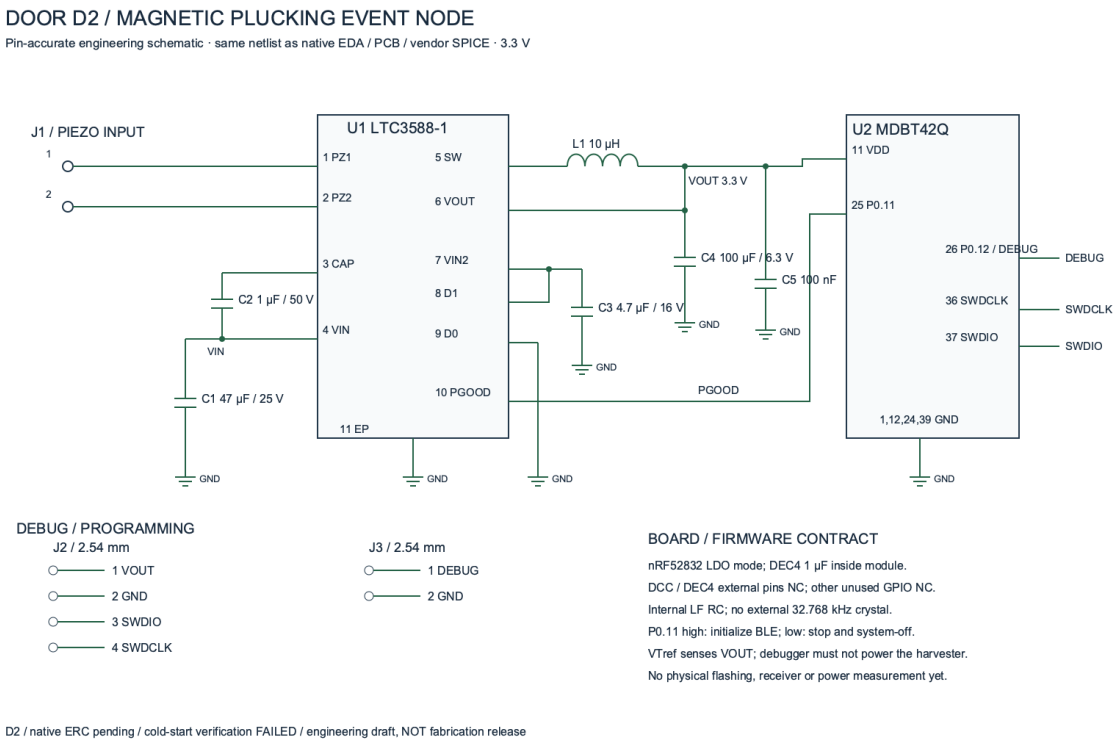


图 8: 与 PCB 同一器件引脚网表生成的电源和无线模块原理图；原生 EDA 读取尚未验收。

标号	选型及作用
U1	LTC3588EMSE-1，压电整流、欠压锁定与降压稳压；输出设置 3.3 V
U2	MDBT42Q-512KV2，nRF52832 模块，LDO 方式
C1	47 μF / 25 V / 1210，整流后输入储能
C2	1 μF / 50 V / 0603，CAP 与 VIN 之间
C3	4.7 μF / 16 V / 0603，VIN2 旁路
C4	100 μF / 6.3 V / 1206，输出储能
C5	100 nF / 50 V / 0603，模块供电旁路
L1	10 μH，SRN4018-100M
J1/J2/J3	压电输入 / SWD 与电压参考 / 时序调试

器件、封装和焊盘来自缓存的立创器件库导出，原始导出和引脚映射一并保留。MDBT42Q 的 DEC4 电容按模块手册内部配置处理，不另造不存在的外部连接。PGOOD 接模块 25 脚（P0.11），时序信号接 26 脚（P0.12）；不能将模块焊盘编号直接当作 MCU 端口编号。

8 厂商 SPICE 电源验证

使用 ADI LTC3588-1 宏模型及真实 LTspice 执行。降阶结果的 $\theta\dot{q}$ 作为 PWL 电流源，与 C_p 并联后驱动 PZ1/PZ2。所有储能电容初始为 0 V，PGOOD 超过 2.5 V 才启用 2 mA 负载。

这是单向耦合：运动来自电阻负载下的降阶计算，尚未把非线性 PMIC 的电压反作用送回机械方程。因此不能用该模型证明产品性能；当前负结果仍清楚显示所选运动输入无法给电源启动。

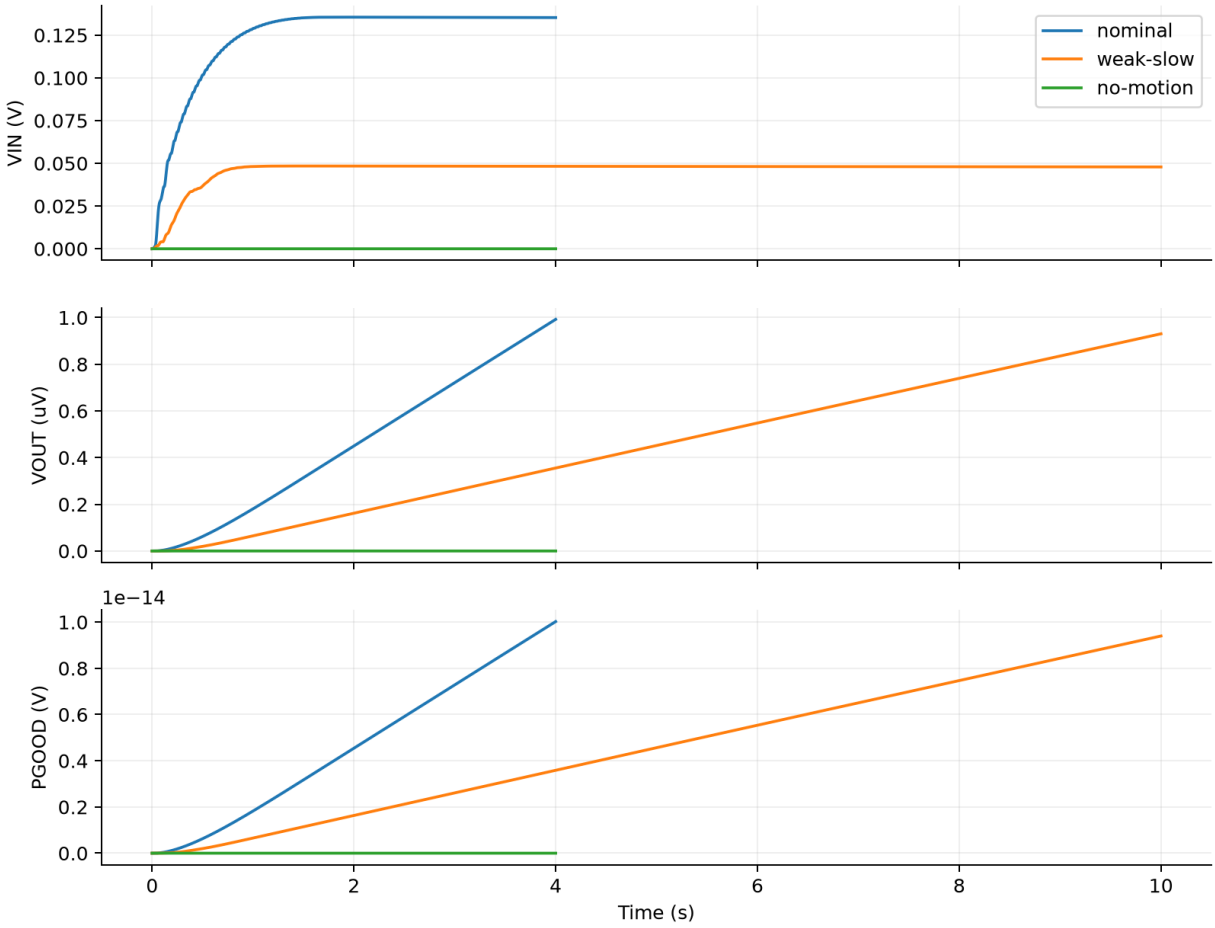


图 9: 从 LTspice 原始二进制 RAW 直接转换的曲线。VOUT 图以微伏为单位，避免把极小残值画成正常供电。

工况	VIN 峰值/V	VOUT 峰值	冷启动
正常	0.135532	约 0.9923 μV	未发生
慢开	0.048421	近零	未发生
无动作	0	0	未发生

正常工况负载能量为 0。仅把 47 μF 输入电容从 0 充到 5.05 V 就需约 599 μJ，且还未计入损耗与输出储能。这个数量级比较不能代替厂商模型，但能解释为何当前连续缓慢磁加载难以形成有效冷启动。

9 PCB 布局、布线与制造边界

板外形为 80×30 mm，两个铜层，4 个安装孔。无线模块位于板端，天线和内凹禁布区域同时施加于两层；供电转换部分在另一端，SWD 连接器与调试接口可达。

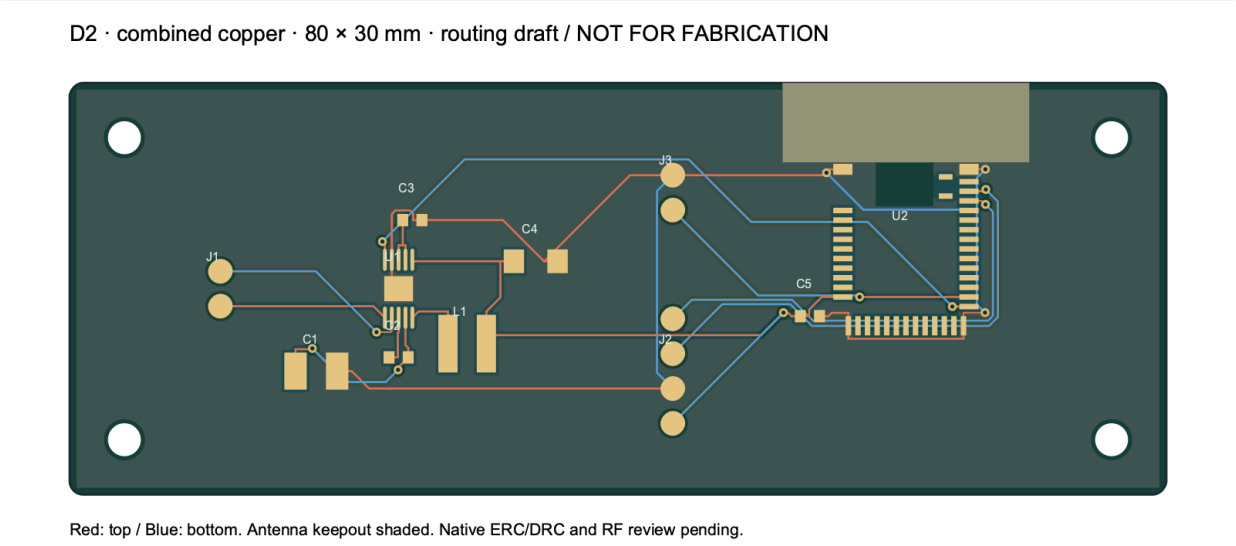


图 10: 实际布线和铜区预览。68 段铜线、12 个过孔、72 个焊盘；图像由同一布线结果生成。

Freerouter 路由后，独立连通性及几何检查均报告 0 断网、0 间距违规和 0 天线禁布冲突。最小铜间距约 0.1669 mm；最窄颈缩线宽约 0.1142 mm，而一般线宽目标为 0.1524 mm。是否接受颈缩线宽，需要选定板厂规则并再次检查。地铜区以真实边界、避让及连接检查生成，不是绘制一个装饰性矩形。

已经输出专业版 EDA 草案文件、路由交换文件以及铜层、板框、钻孔 Gerber 草案。原生 EDA 导入调用返回空结果，尚未完成原生 ERC/DRC 和可编辑性回读；不把独立几何检查写成原生 DRC。阻焊、钢网、板层叠构、制造公差与订单前审查尚未闭合，因此本包标记为工程样机草案，未释放制造。

10 MCU 固件与时序

目标为 nRF52832 xxAA，S132 3.0.0，固定 SDK 源提交。固件使用内部低频 RC 时钟，不依赖外置 32.768 kHz 晶体。PGOOD 低时进入 SYSTEMOFF，配置上升唤醒；可用供电时发送非连接 BLE 广播；检测到 PGOOD 低后停止发送并休眠。欠压期间 MCU 行为仍需实物棕断测试。

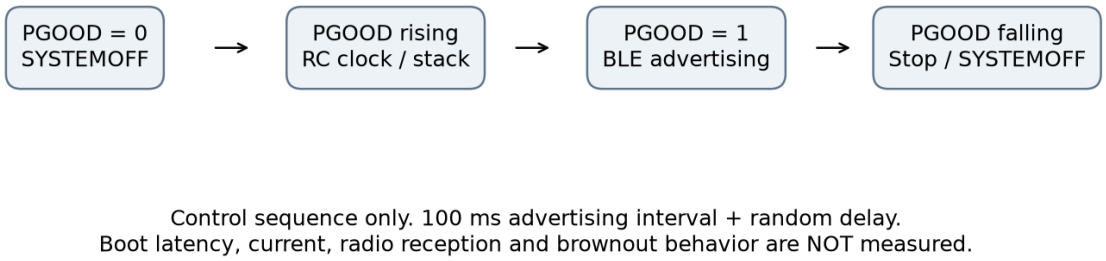


图 11: 由控制分支整理的时序关系，横向位置不代表实测耗时。

项目	已验证数据
应用 Flash	4896 bytes，不含 SoftDevice
静态 RAM	180 bytes，不含动态峰值与协议栈占用
栈 / 堆保留	各 8192 bytes，合计预留 RAM 16564 bytes
应用区域	Flash 397312 bytes，RAM 52496 bytes
广播	100 ms 配置间隔，叠加协议随机延迟；0 dBm
载荷	厂商号 0xffff，BFD2 + 8 字节 FICR 器件 ID
重复构建	clean 后 HEX/BIN 哈希一致
烧录	已生成烧录脚本和 dry-run；未连接目标板、未实际执行
接收	解码器与合成协议样例检查；没有真实无线接收数据

事件按照接收端静默间隔分组，不能从 BFD2 载荷可靠推断开门/关门方向，也没有跨重启持久事件编号。不同包数不能自动解释为方向差异。论文给出的 72.70、68.47、60.37 μJ 阶段能量只是对照资料，绝不是这个二进制的功耗测量。

11 验收矩阵与下一次修订

检查	状态	证据与限制
CAD 实体回读	通过	22 个有效实体，原生 STEP/BREP
刚体安装运动	通过	66 个角度样本，非连续扫掠
梁静载网格	部分通过	两级位移、应力差异小；未覆盖疲劳和非线性
磁力网格	未通过	最大向量变化 7.65% > 5%
降阶时间步	通过	1 ms / 0.5 ms 能量差异远小于 5%
负对照	通过	无动作、无磁体输出为零
单次开门冷启动	未通过	PGOOD 未拉高，负载无供能
PCB 独立检查	通过	连通、间距、天线禁布几何检查
原生 EDA ERC/DRC	未执行	导入返回空结果，未取得原生报告
固件目标编译	通过	ELF/BIN/HEX，重复构建一致

实物功耗/时延/射频	未执行	未连接硬件
完整产品任务	未通过	冷启动及硬件证据尚缺

下一版应首先改变机械激励形成方式，使储存的机械能更快释放，或选用经验证的阻抗匹配和储能方案。单纯把磁体继续靠近会重新进入大挠度区，必须先建立几何非线性或接触/脱离模型，再修改间距。电源与机械应双向耦合，不能用预先计算的电流波形绕过反作用。随后完成磁网格收敛、原生 PCB 回读与制造规则、最终硬件电压/电流/接收同步测量。

更好的渲染不改变上述判定。D2 的用途是提供可打开、可修改、可复算和可定位失败的数字工程基线。

12 文件、复算与来源

工程包包含 requirements、mechanical-D2、animation-D2、simulation、circuit、pcb、firmware-D2-release、report 与 reproduction。文件清单记录 SHA-256；产品页按机械、仿真、电路、PCB、固件和报告展示同一版本。

报告源为 `report/door-case.tex`，使用 XeLaTeX 连续编译两次生成本 PDF，图像来自同一目录的 `figures`。替换设计参数后必须重新计算，不能仅替换报告中的数字。完整复算依赖 CadQuery、Blender、COMSOL 6.2、LTspice、立创专业版、ARM 工具链及各自许可证；原生文件保留，未把商业求解器打包分发。

参考资料：

1. ViPSN-Pluck: A Transient-Motion-Powered Motion Detector；作者公开 PDF。
2. A unified system-level simulation framework for energy harvesting, management, and application in plucking piezoelectric energy harvesters；DOI 10.1088/1361-665X/add8d4。
3. Cross-Domain Simulation of a Plucking Piezoelectric Energy Harvester: Coupling Dynamics, Power Management, and Application；DOI 10.1115/1.4071279。
4. ADI LTC3588-1 数据手册及 LTspice 宏模型；官方 35881fc 数据手册。
5. Raytac MDBT42Q 模块资料 Version O；器件供应链公开 PDF，天线布局与 LDO 说明按文档校对。

第三方资料和 Nordic SDK / SoftDevice 保留原许可，仅作为工程参考与复算依赖，不重新授权。实际来源、下载记录及哈希保存在 sources 与各求解回执中。

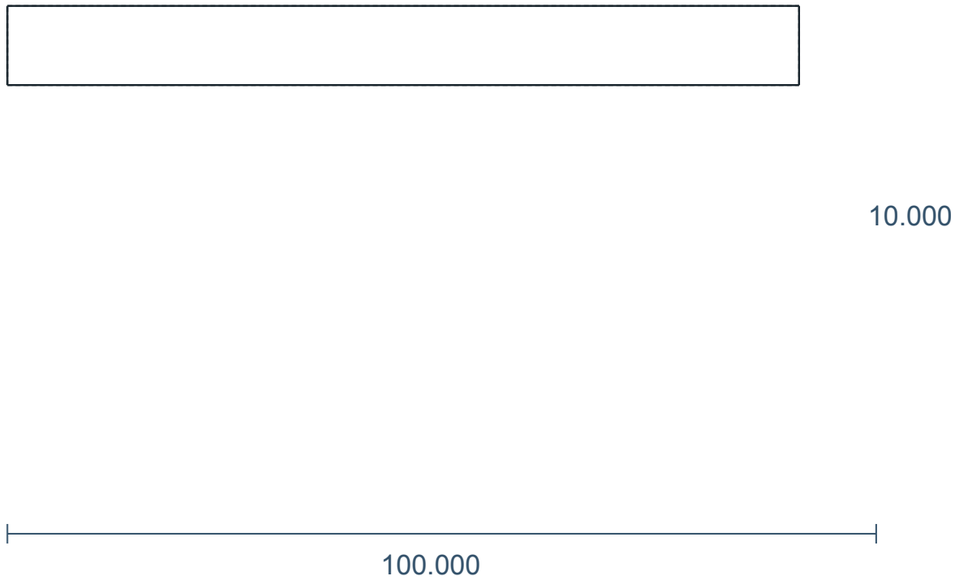
A 22 张原生 CAD 投影图

以下图纸来自同一 D2 几何，当前含包络尺寸。加工所需的功能尺寸、孔位公差、配合和表面要求仍需生产制图补齐。

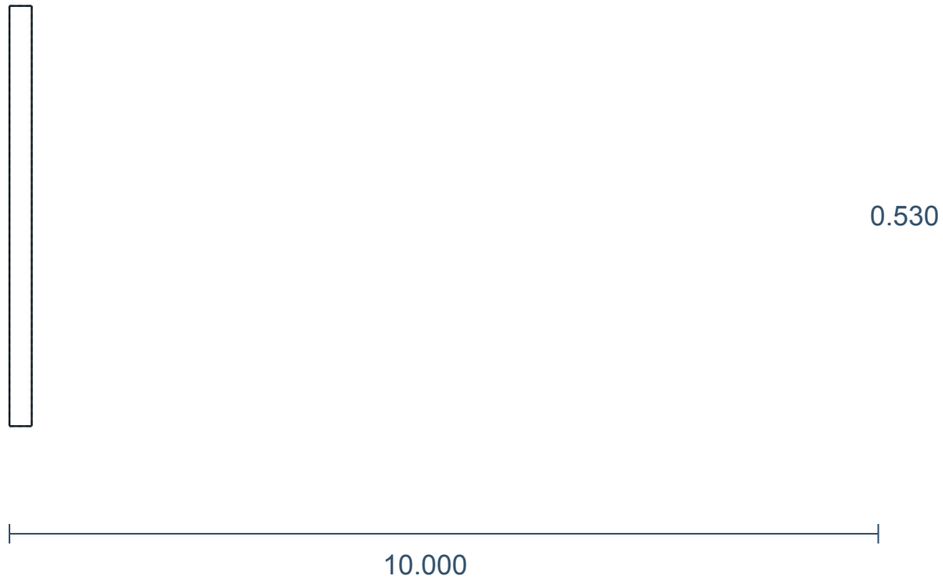
BF DOOR / D2 / beam

Units mm | third-angle views | prototype drawing | projection from STEP geometry

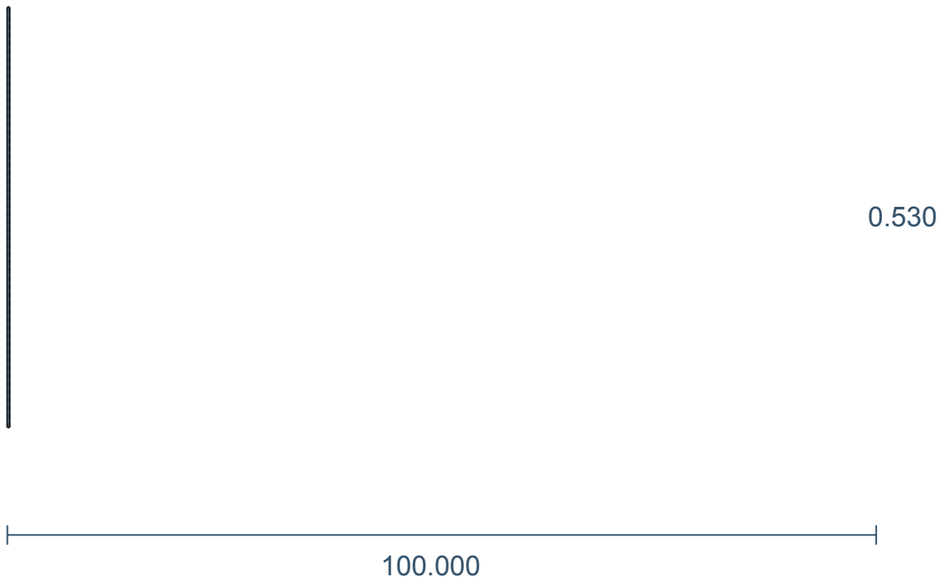
TOP



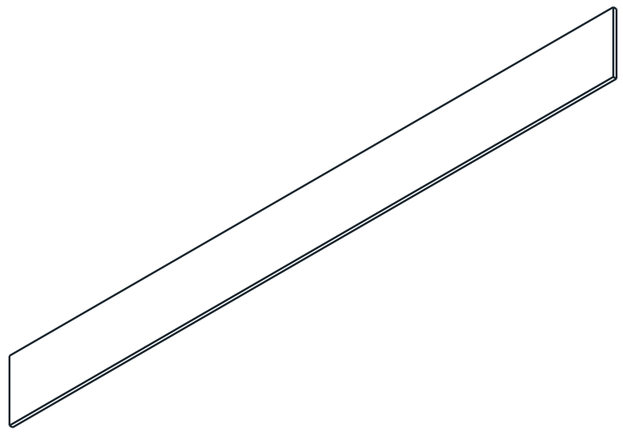
RIGHT



FRONT



ISOMETRIC



Material: copper | envelope 100.000 x 10.000 x 0.530 mm

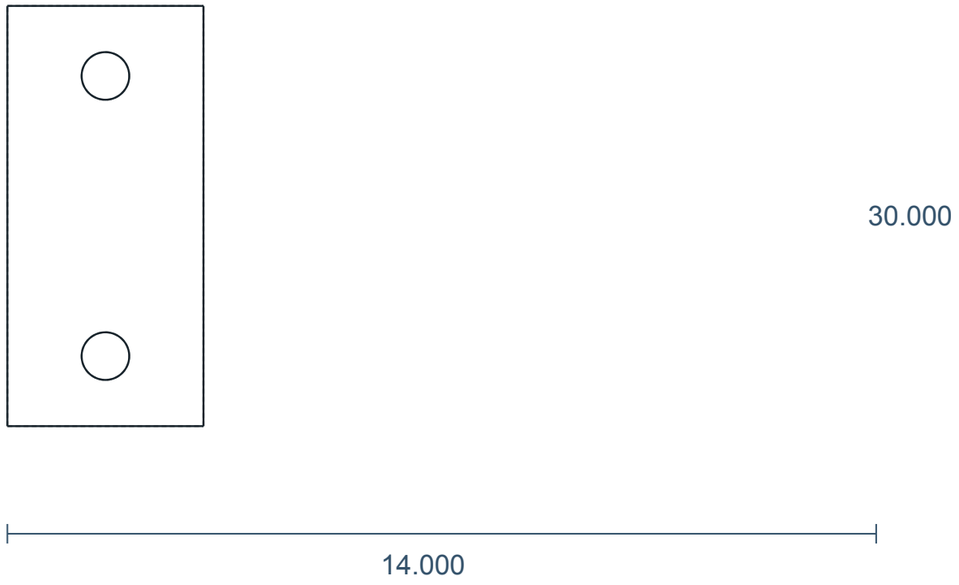
Plastic general tolerance +/-0.25 mm; metal/ceramic tolerance requires supplier drawing.

Hole positions and boolean features: input.json. No IP, fatigue, thread-strength or fabrication-release certification.

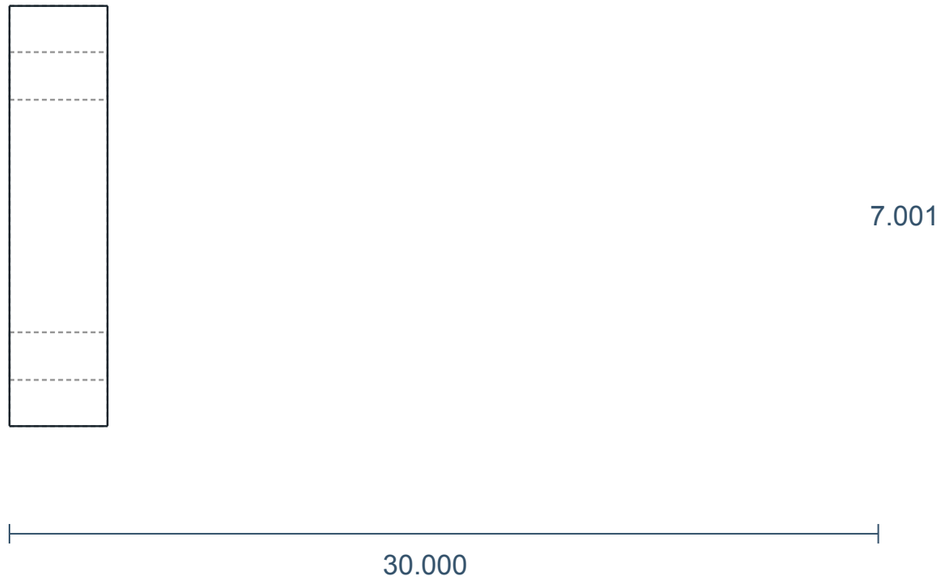
BF DOOR / D2 / clamp_lower

Units mm | third-angle views | prototype drawing | projection from STEP geometry

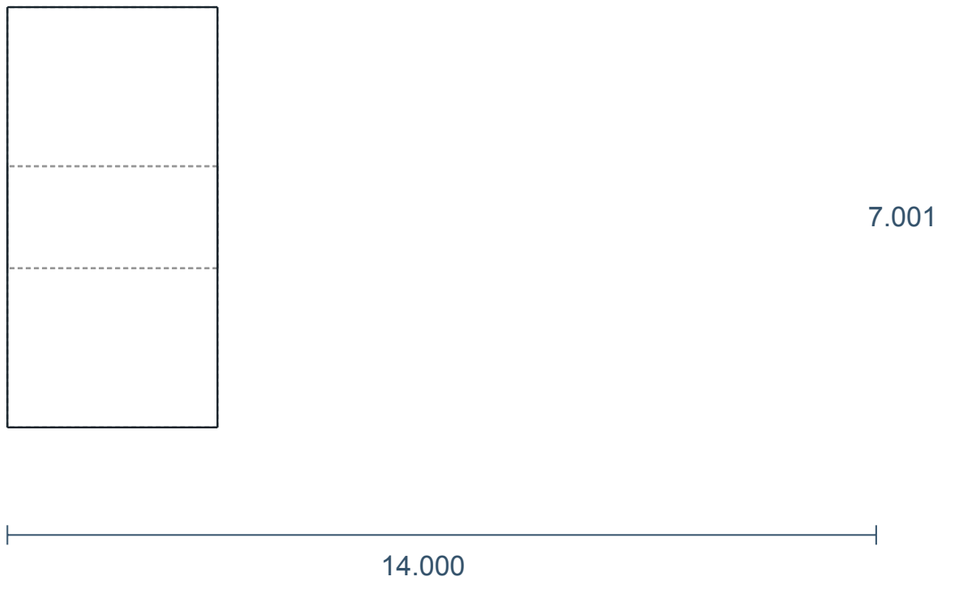
TOP



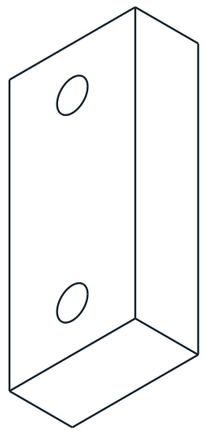
RIGHT



FRONT



ISOMETRIC



Material: PETG | envelope 14.000 x 30.000 x 7.001 mm

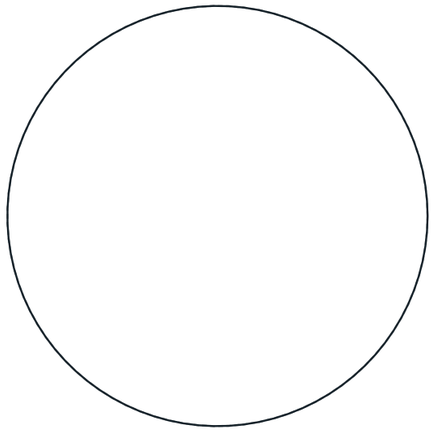
Plastic general tolerance +/-0.25 mm; metal/ceramic tolerance requires supplier drawing.

Hole positions and boolean features: input.json. No IP, fatigue, thread-strength or fabrication-release certification.

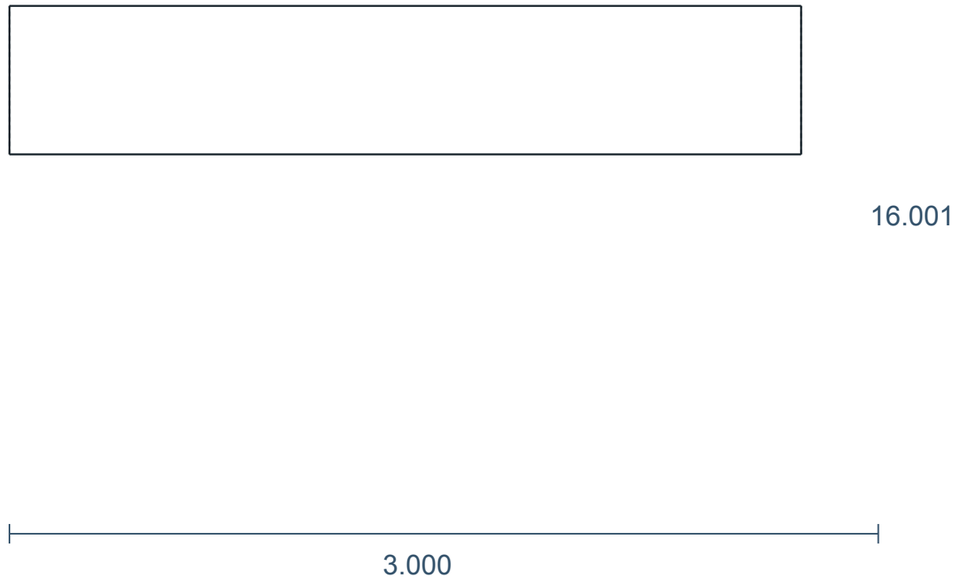
BF DOOR / D2 / clamp_screw_0

Units mm | third-angle views | prototype drawing | projection from STEP geometry

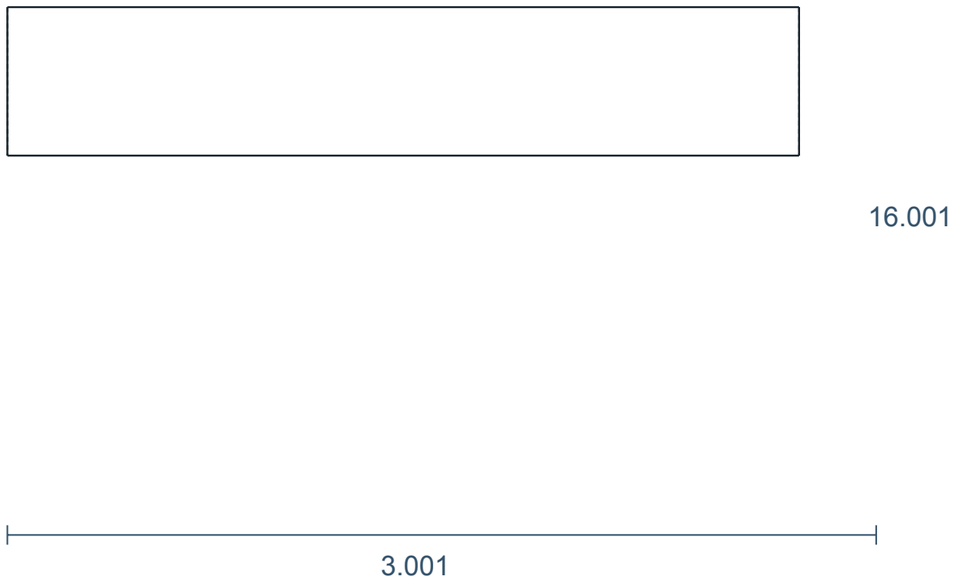
TOP



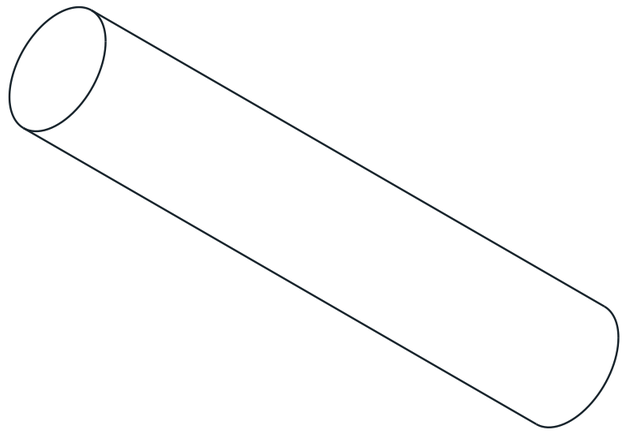
RIGHT



FRONT



ISOMETRIC



Material: M3 nylon | envelope 3.001 x 3.000 x 16.001 mm

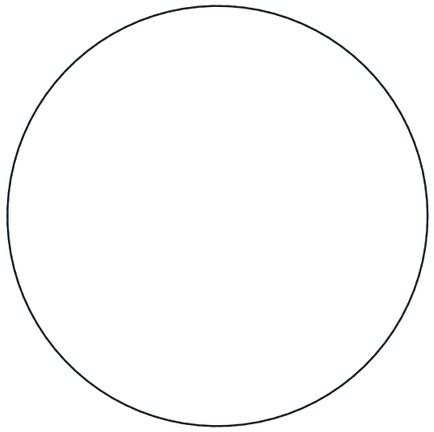
Plastic general tolerance +/-0.25 mm; metal/ceramic tolerance requires supplier drawing.

Hole positions and boolean features: input.json. No IP, fatigue, thread-strength or fabrication-release certification.

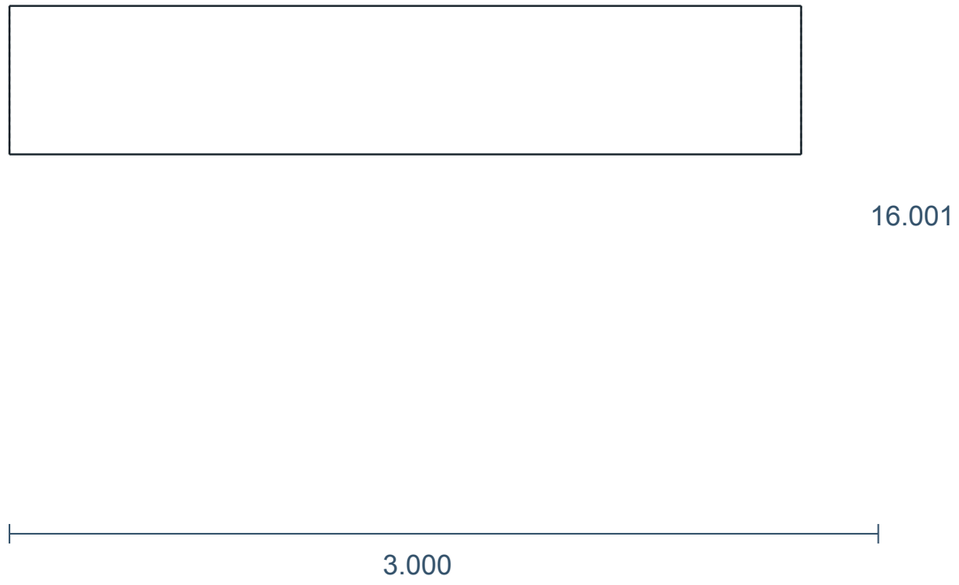
BF DOOR / D2 / clamp_screw_1

Units mm | third-angle views | prototype drawing | projection from STEP geometry

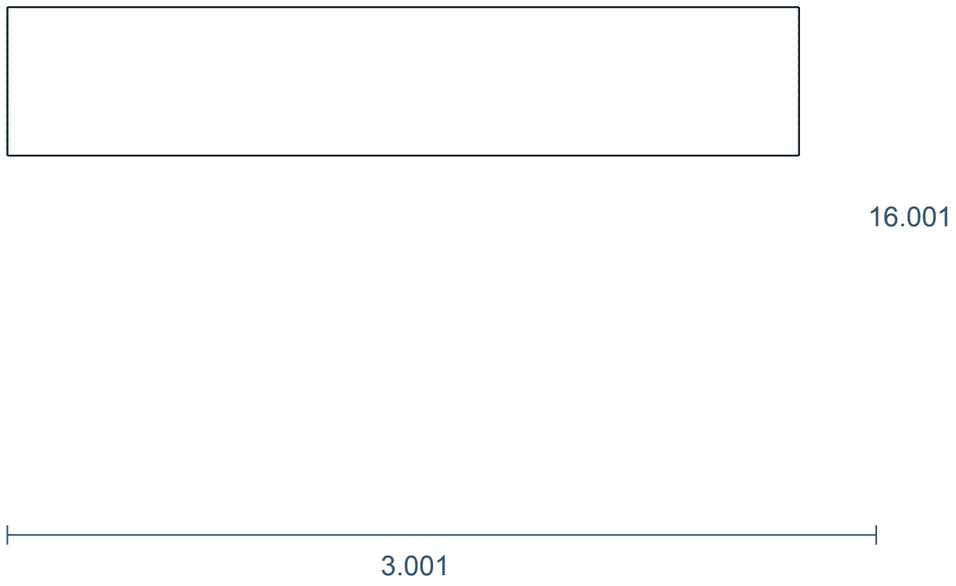
TOP



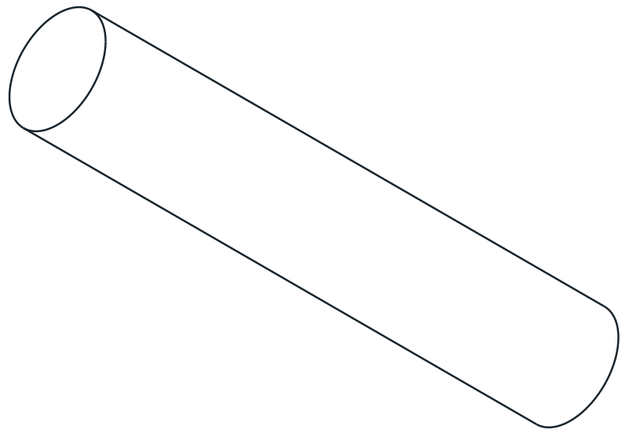
RIGHT



FRONT



ISOMETRIC



Material: M3 nylon | envelope 3.001 x 3.000 x 16.001 mm

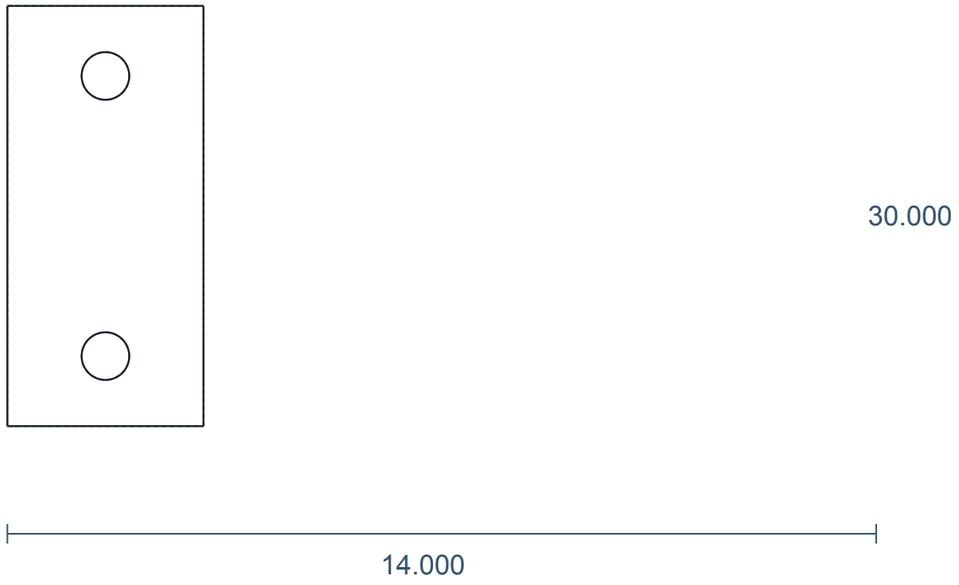
Plastic general tolerance +/-0.25 mm; metal/ceramic tolerance requires supplier drawing.

Hole positions and boolean features: input.json. No IP, fatigue, thread-strength or fabrication-release certification.

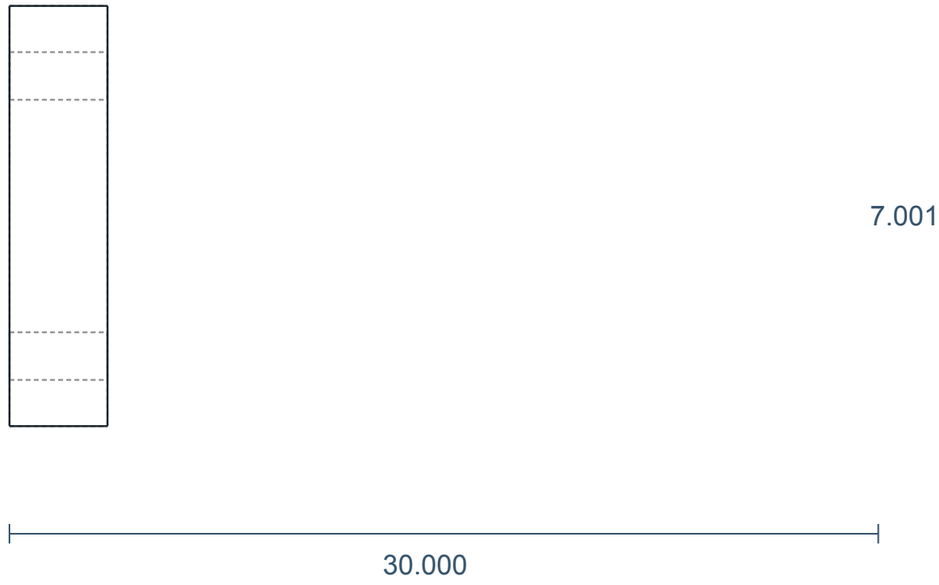
BF DOOR / D2 / clamp_upper

Units mm | third-angle views | prototype drawing | projection from STEP geometry

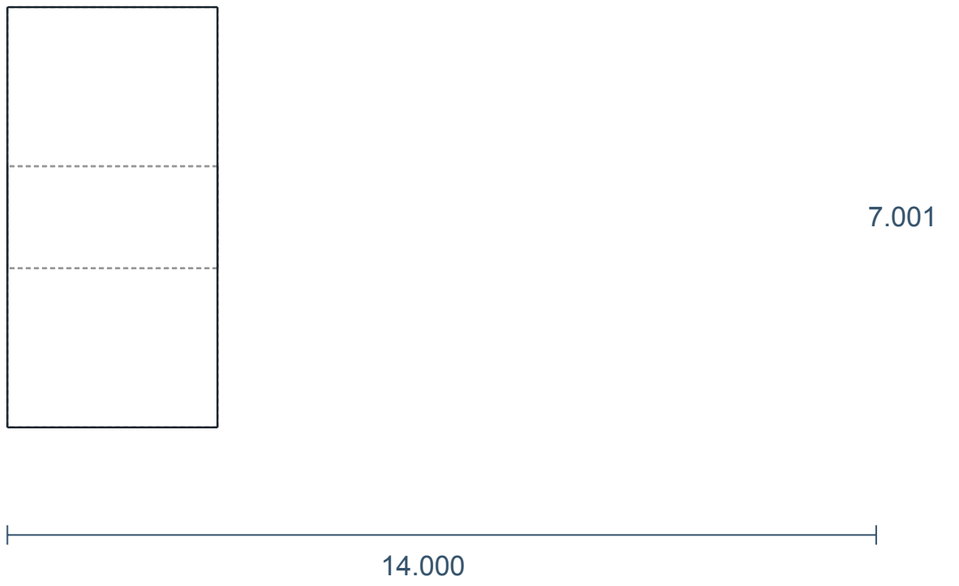
TOP



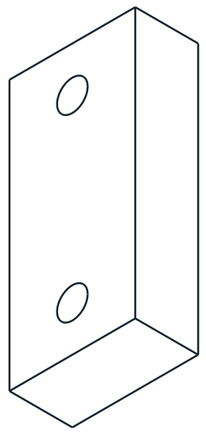
RIGHT



FRONT



ISOMETRIC



Material: PETG | envelope 14.000 x 30.000 x 7.001 mm

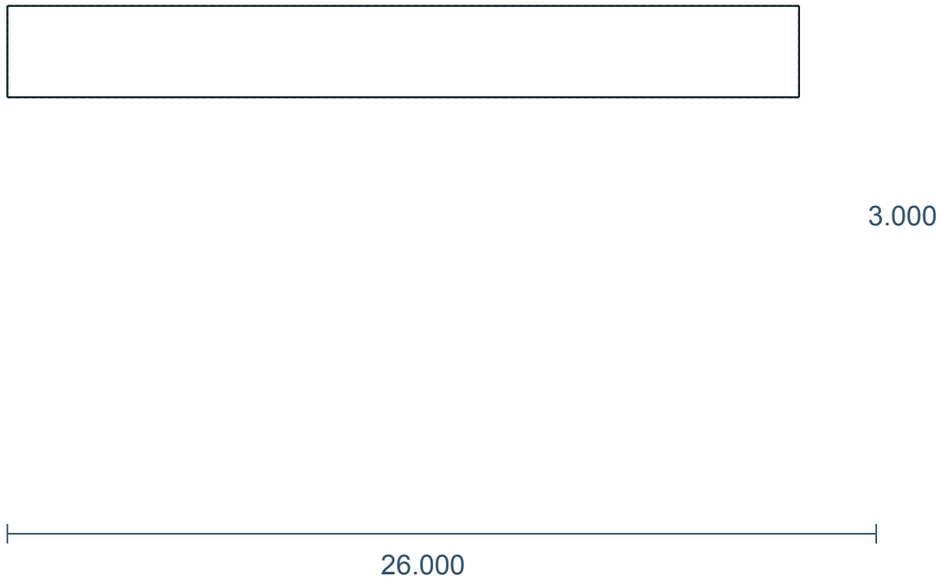
Plastic general tolerance +/-0.25 mm; metal/ceramic tolerance requires supplier drawing.

Hole positions and boolean features: input.json. No IP, fatigue, thread-strength or fabrication-release certification.

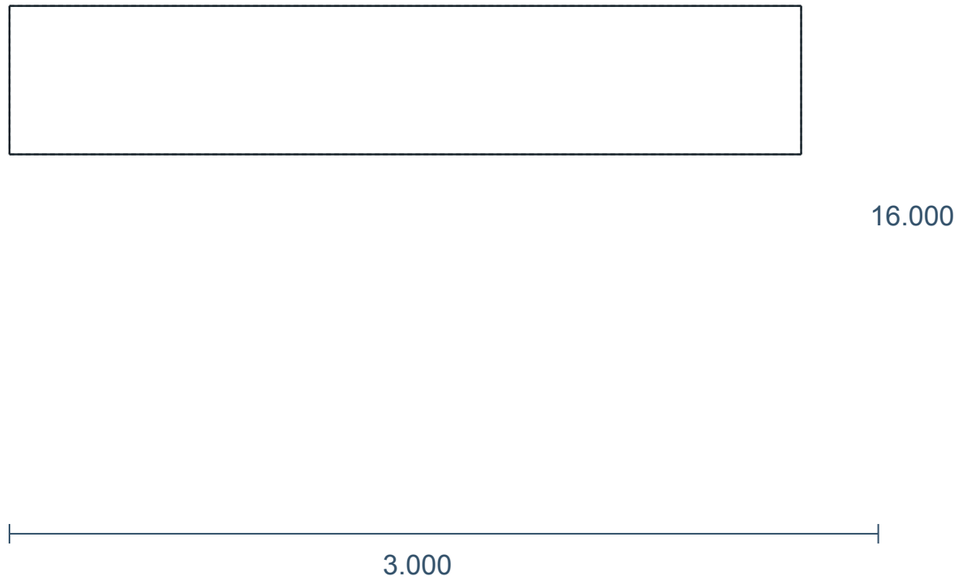
BF DOOR / D2 / driver_bracket_shelf

Units mm | third-angle views | prototype drawing | projection from STEP geometry

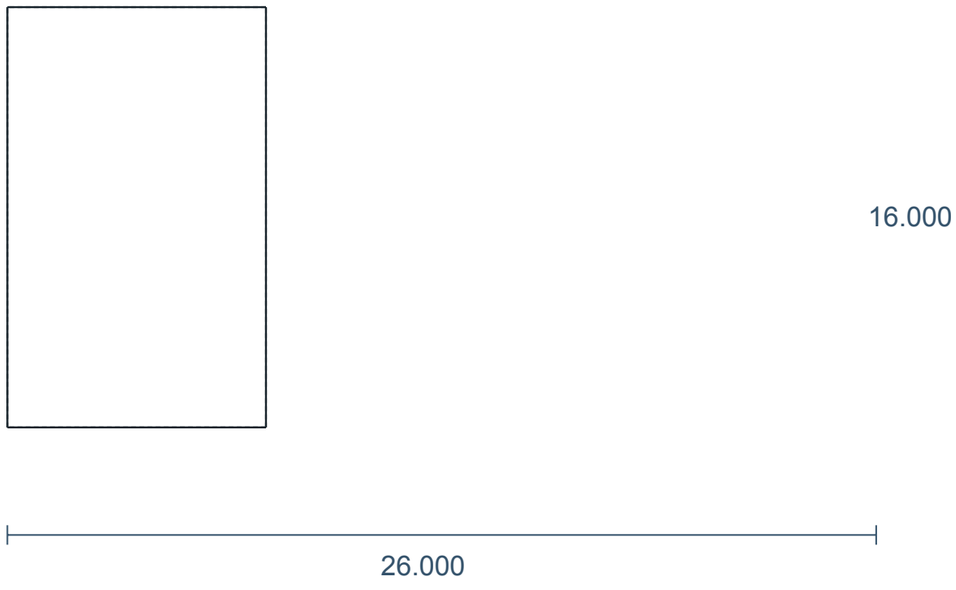
TOP



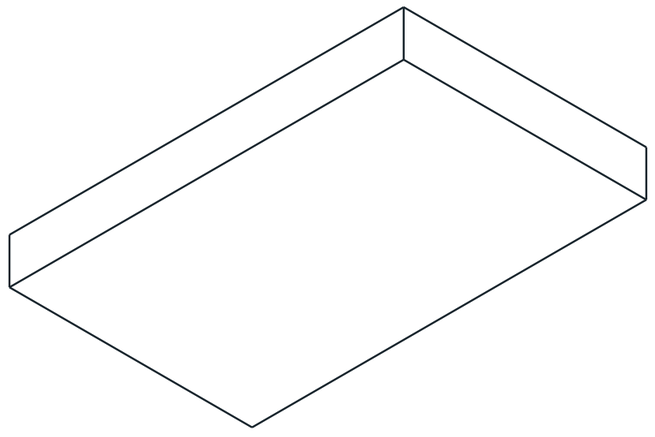
RIGHT



FRONT



ISOMETRIC



Material: PETG prototype | envelope 26.000 x 3.000 x 16.000 mm

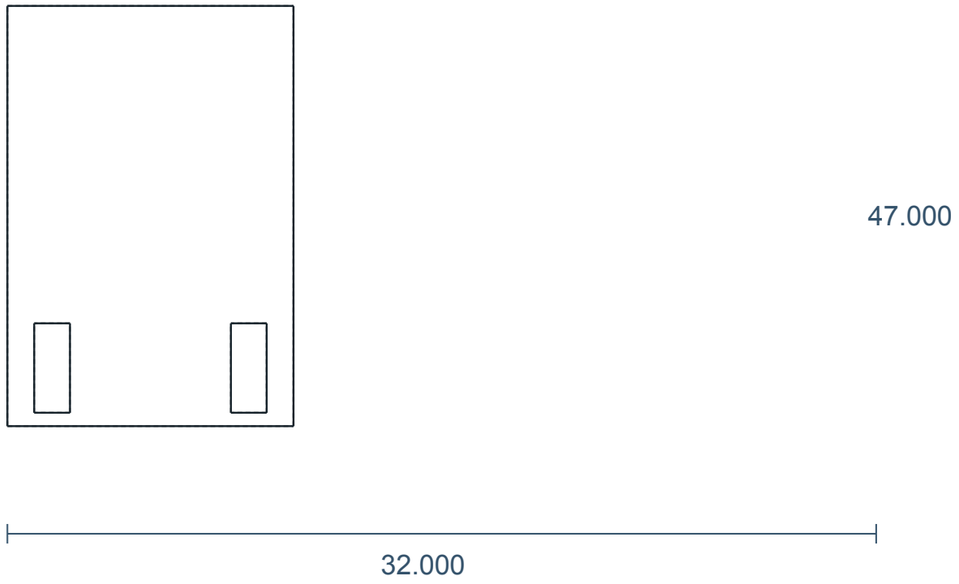
Plastic general tolerance +/-0.25 mm; metal/ceramic tolerance requires supplier drawing.

Hole positions and boolean features: input.json. No IP, fatigue, thread-strength or fabrication-release certification.

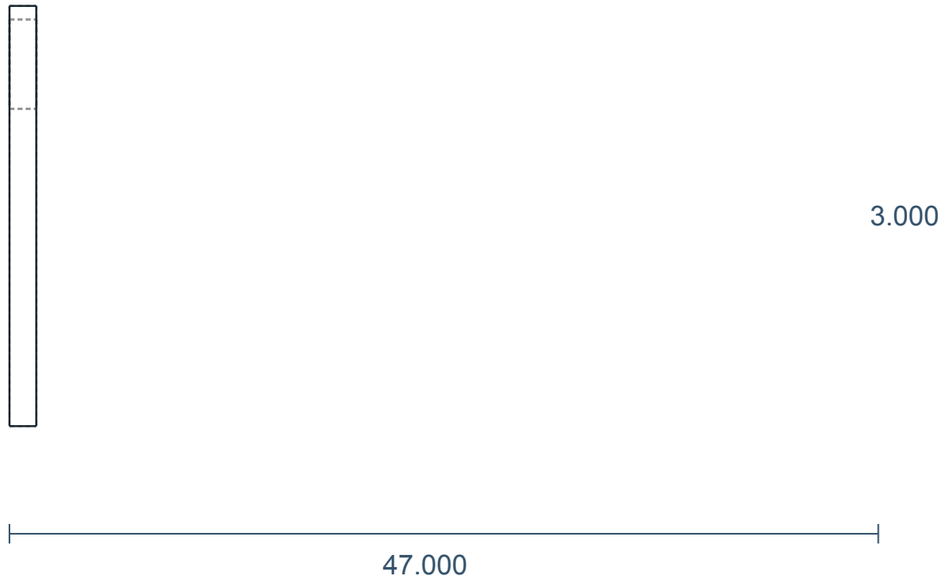
BF DOOR / D2 / driver_bracket_vertical

Units mm | third-angle views | prototype drawing | projection from STEP geometry

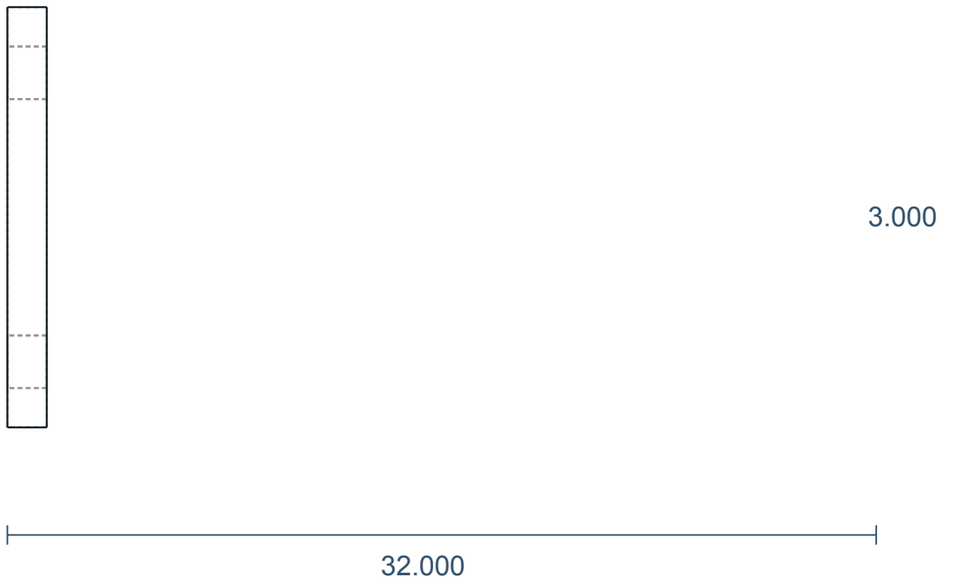
TOP



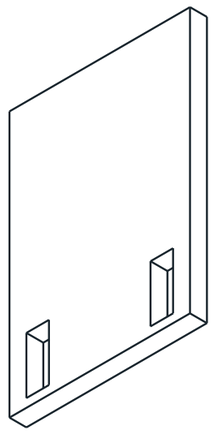
RIGHT



FRONT



ISOMETRIC



Material: PETG prototype | envelope 32.000 x 47.000 x 3.000 mm

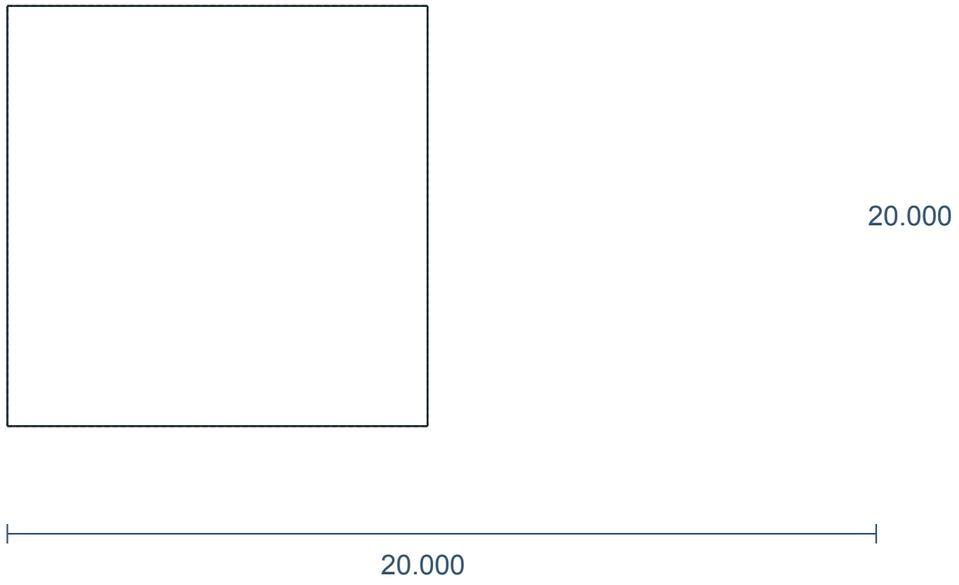
Plastic general tolerance +/-0.25 mm; metal/ceramic tolerance requires supplier drawing.

Hole positions and boolean features: input.json. No IP, fatigue, thread-strength or fabrication-release certification.

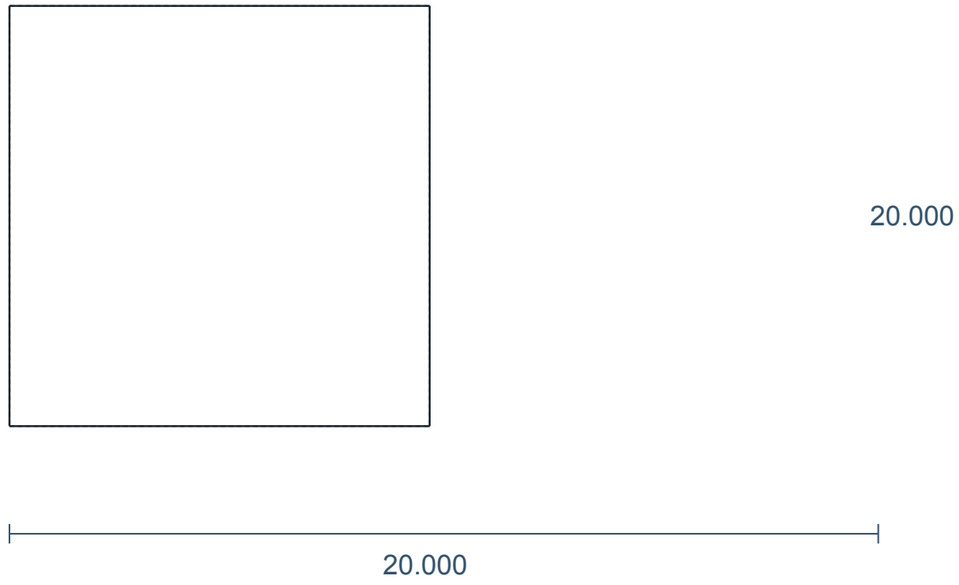
BF DOOR / D2 / driver_magnet

Units mm | third-angle views | prototype drawing | projection from STEP geometry

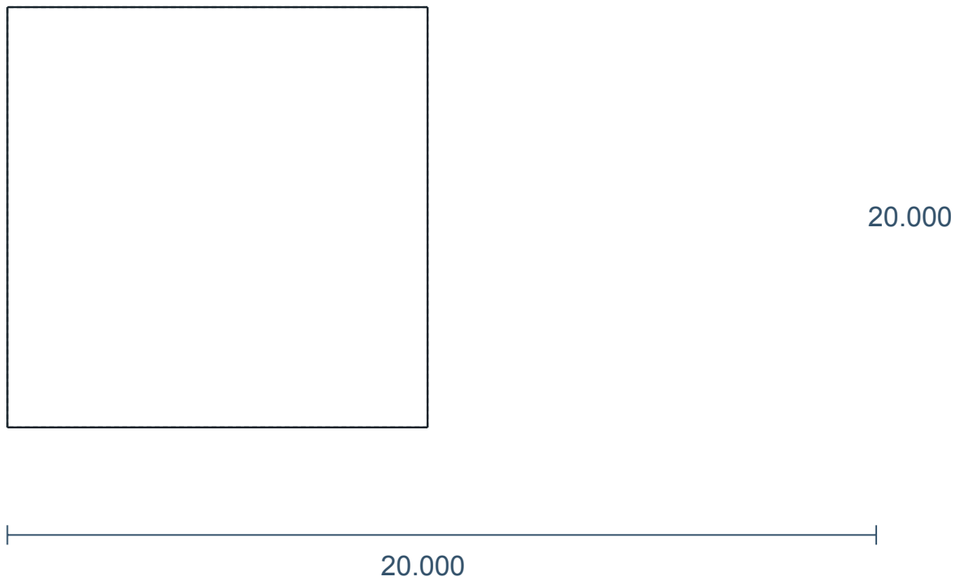
TOP



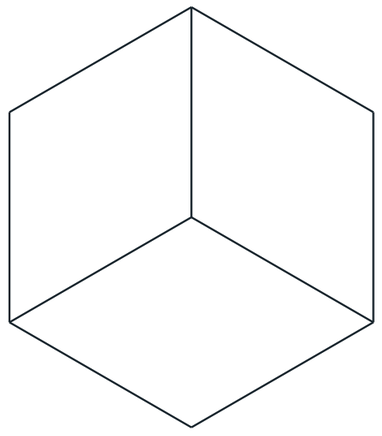
RIGHT



FRONT



ISOMETRIC



Material: NdFeB N42 nominal Br=1.25 T | envelope 20.000 x 20.000 x 20.000 mm

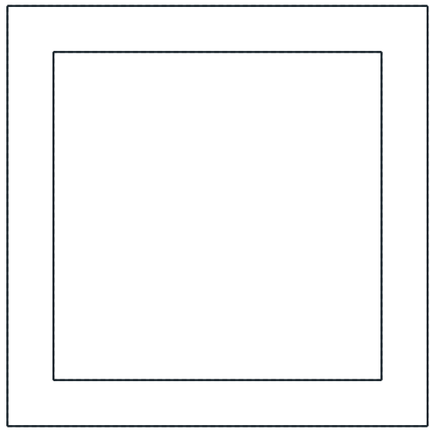
Plastic general tolerance +/-0.25 mm; metal/ceramic tolerance requires supplier drawing.

Hole positions and boolean features: input.json. No IP, fatigue, thread-strength or fabrication-release certification.

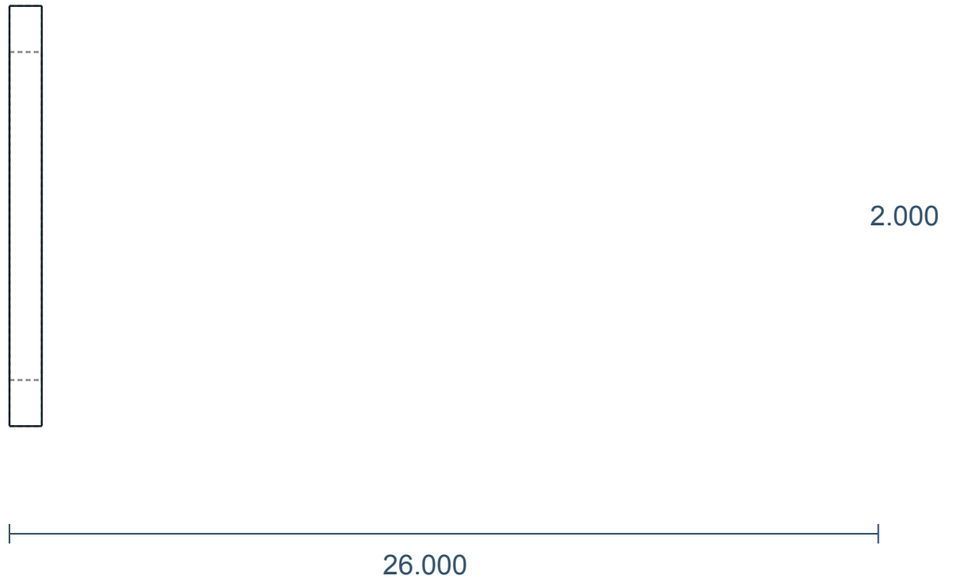
BF DOOR / D2 / driver_retainer

Units mm | third-angle views | prototype drawing | projection from STEP geometry

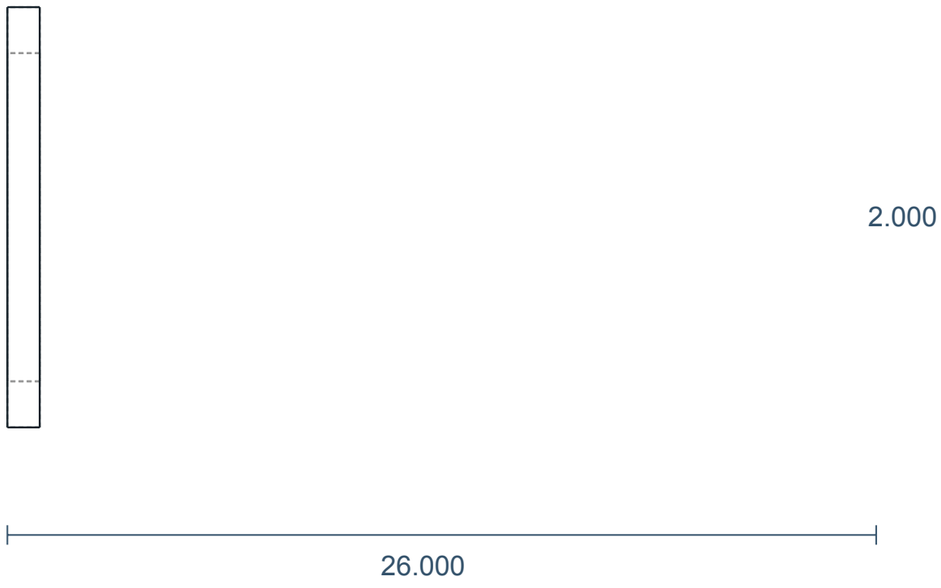
TOP



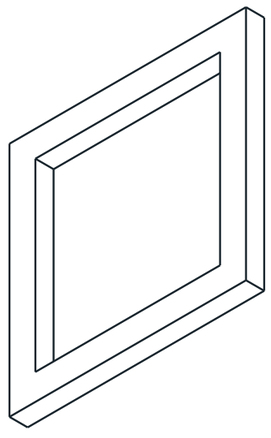
RIGHT



FRONT



ISOMETRIC



Material: PETG prototype | envelope 26.000 x 26.000 x 2.000 mm

Plastic general tolerance +/-0.25 mm; metal/ceramic tolerance requires supplier drawing.

Hole positions and boolean features: input.json. No IP, fatigue, thread-strength or fabrication-release certification.

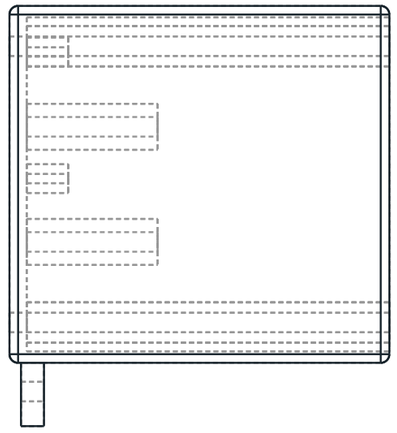
BF DOOR / D2 / housing

Units mm | third-angle views | prototype drawing | projection from STEP geometry

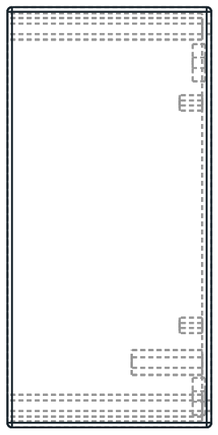
TOP



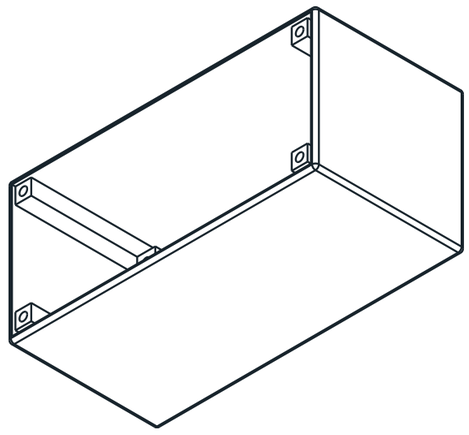
RIGHT



FRONT



ISOMETRIC



Material: PETG, prototype only | envelope 136.003 x 73.002 x 66.003 mm

Plastic general tolerance +/-0.25 mm; metal/ceramic tolerance requires supplier drawing.

Hole positions and boolean features: input.json. No IP, fatigue, thread-strength or fabrication-release certification.

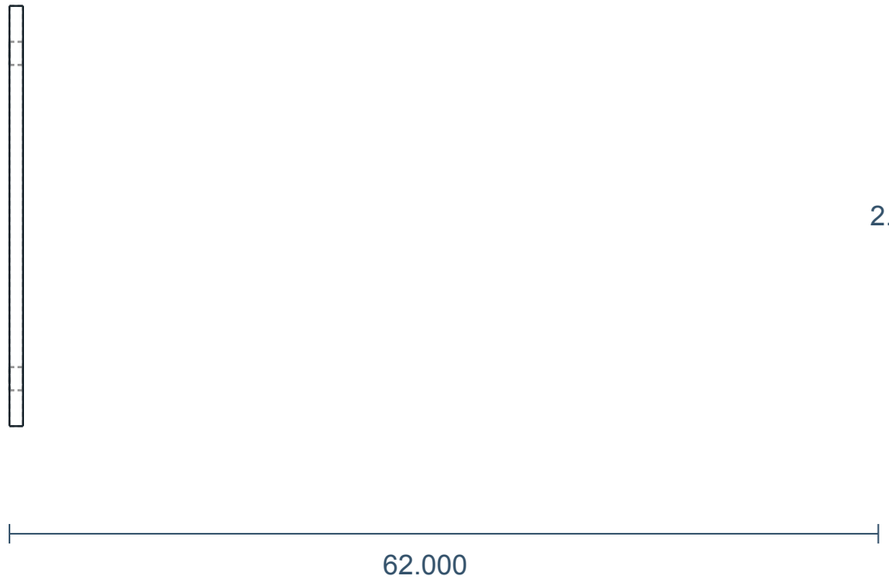
BF DOOR / D2 / lid

Units mm | third-angle views | prototype drawing | projection from STEP geometry

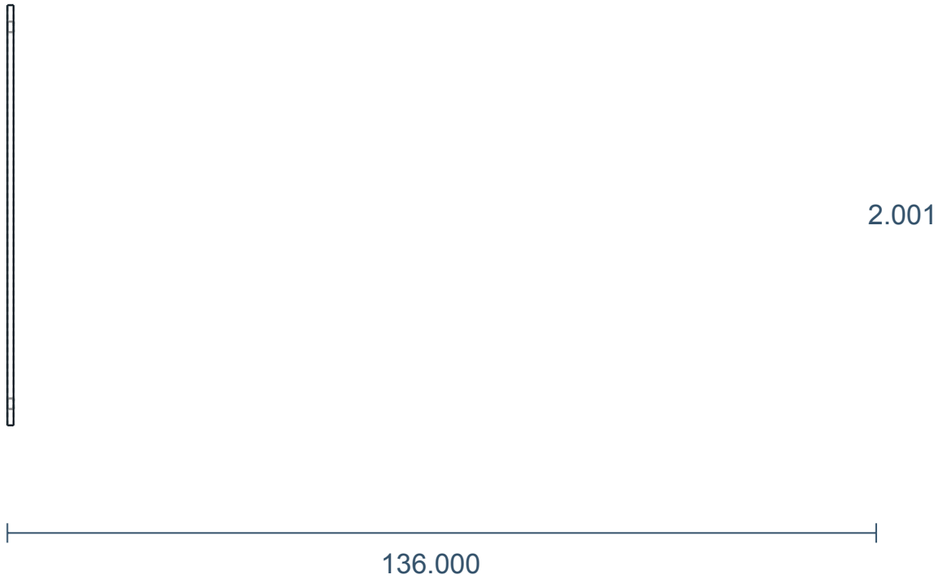
TOP



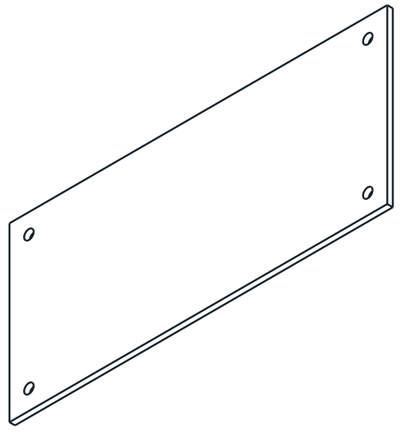
RIGHT



FRONT



ISOMETRIC



Material: clear polycarbonate, assumed | envelope 136.000 x 62.000 x 2.001 mm

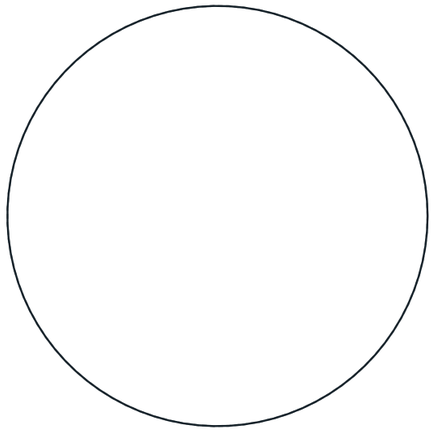
Plastic general tolerance +/-0.25 mm; metal/ceramic tolerance requires supplier drawing.

Hole positions and boolean features: input.json. No IP, fatigue, thread-strength or fabrication-release certification.

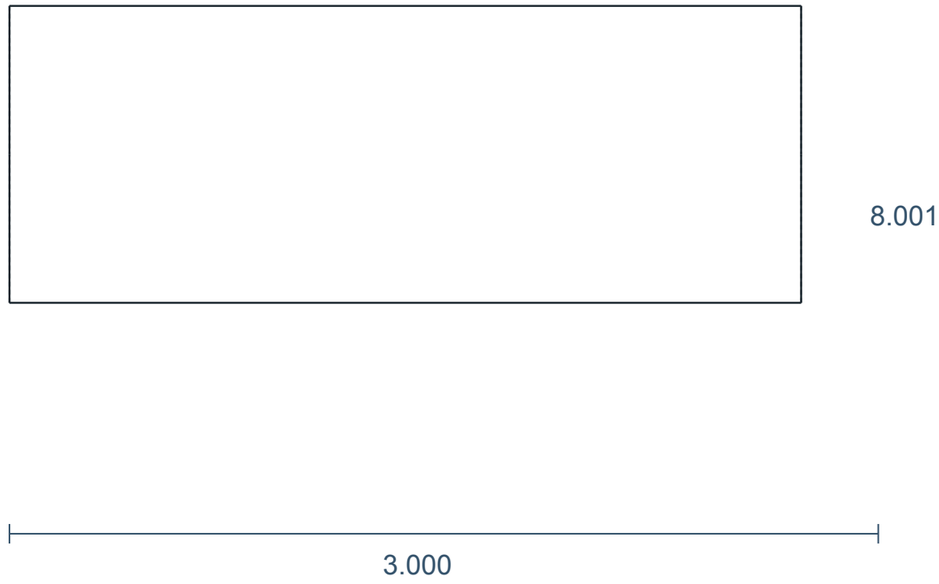
BF DOOR / D2 / lid_fastener_0

Units mm | third-angle views | prototype drawing | projection from STEP geometry

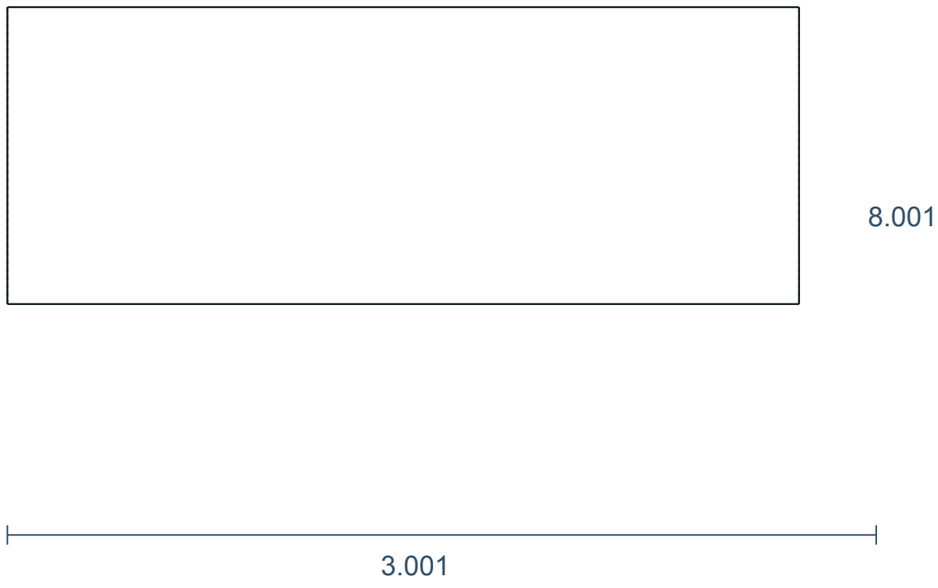
TOP



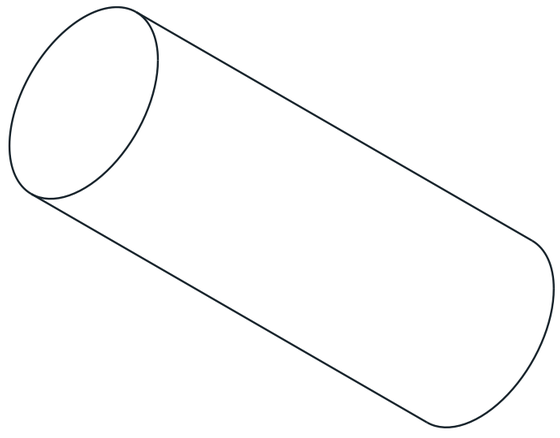
RIGHT



FRONT



ISOMETRIC



Material: M3 nylon | envelope 3.001 x 3.000 x 8.001 mm

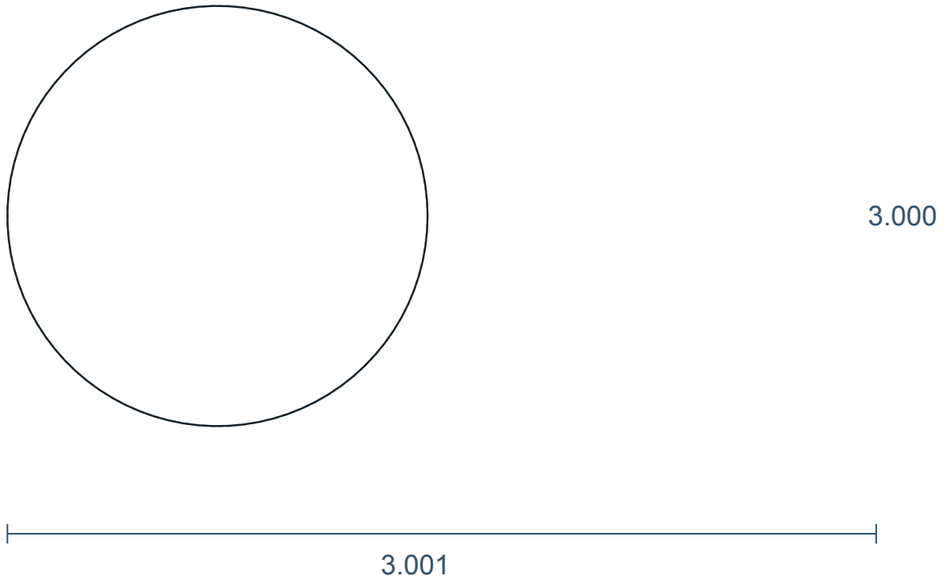
Plastic general tolerance +/-0.25 mm; metal/ceramic tolerance requires supplier drawing.

Hole positions and boolean features: input.json. No IP, fatigue, thread-strength or fabrication-release certification.

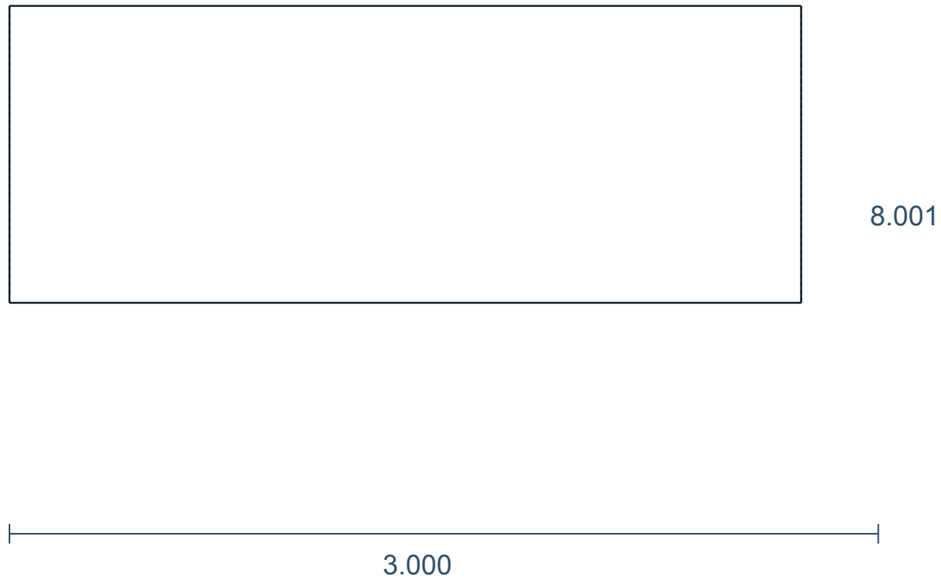
BF DOOR / D2 / lid_fastener_1

Units mm | third-angle views | prototype drawing | projection from STEP geometry

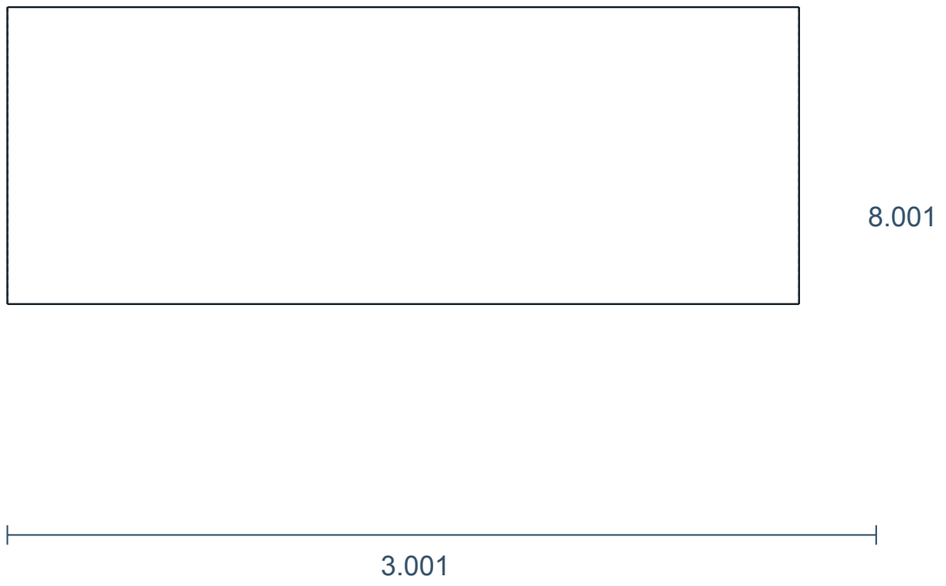
TOP



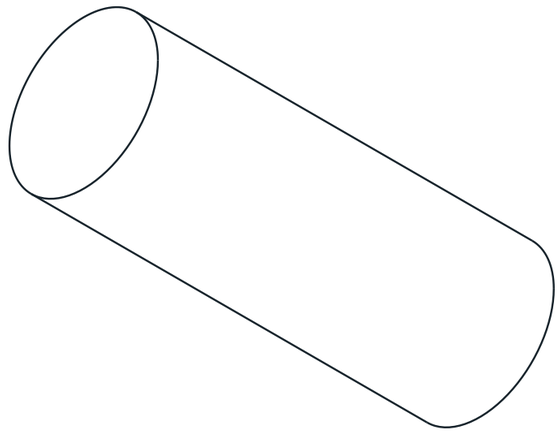
RIGHT



FRONT



ISOMETRIC



Material: M3 nylon | envelope 3.001 x 3.000 x 8.001 mm

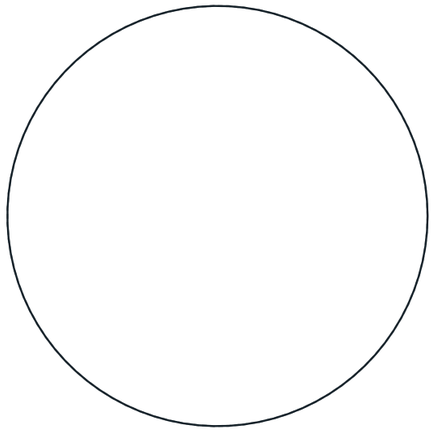
Plastic general tolerance +/-0.25 mm; metal/ceramic tolerance requires supplier drawing.

Hole positions and boolean features: input.json. No IP, fatigue, thread-strength or fabrication-release certification.

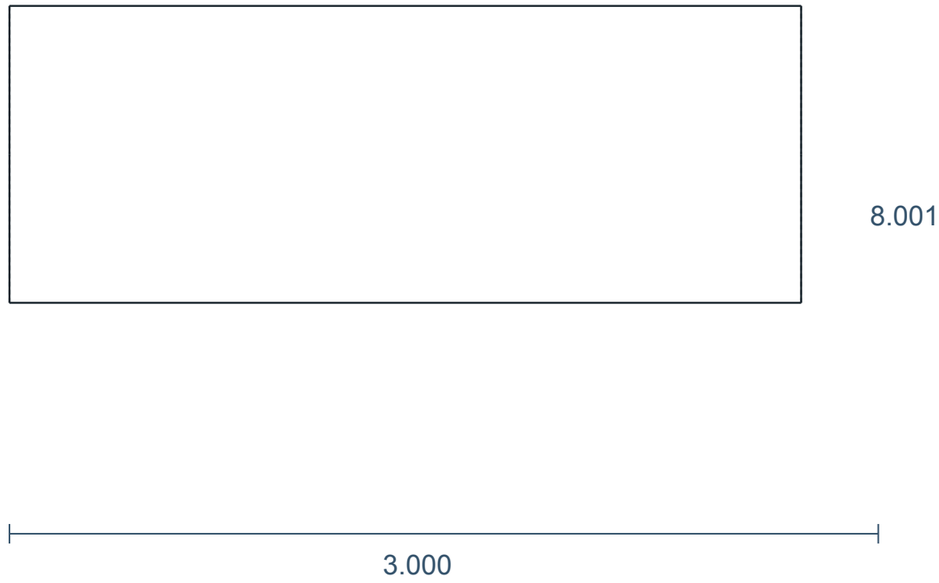
BF DOOR / D2 / lid_fastener_2

Units mm | third-angle views | prototype drawing | projection from STEP geometry

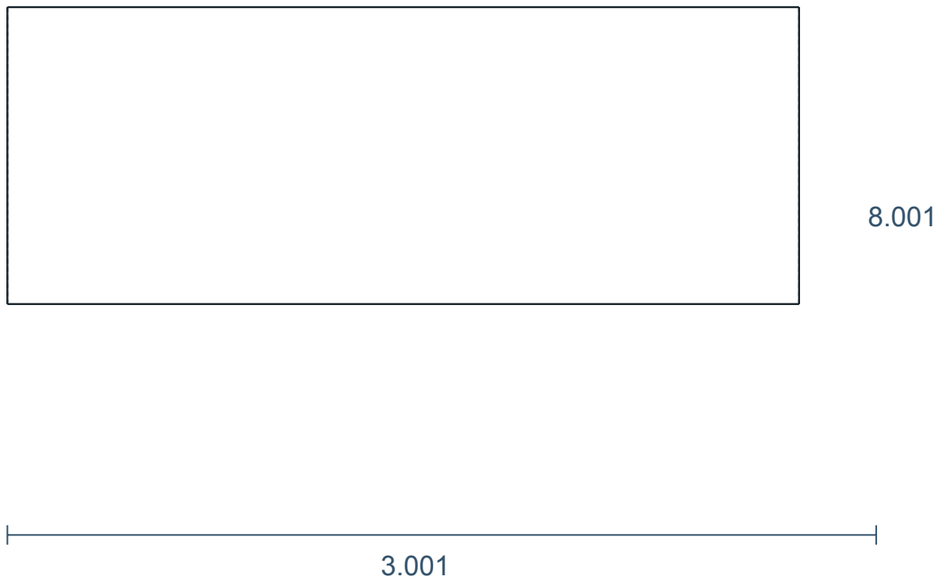
TOP



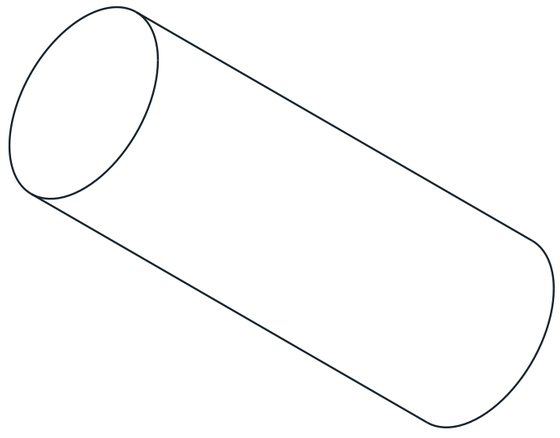
RIGHT



FRONT



ISOMETRIC



Material: M3 nylon | envelope 3.001 x 3.000 x 8.001 mm

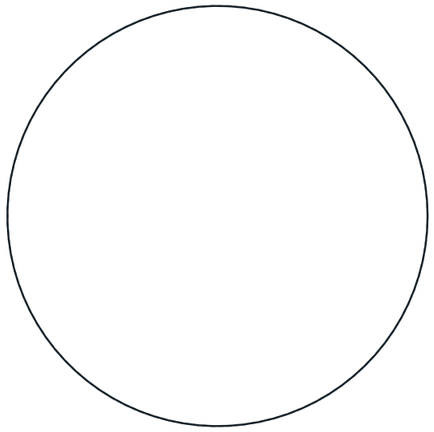
Plastic general tolerance +/-0.25 mm; metal/ceramic tolerance requires supplier drawing.

Hole positions and boolean features: input.json. No IP, fatigue, thread-strength or fabrication-release certification.

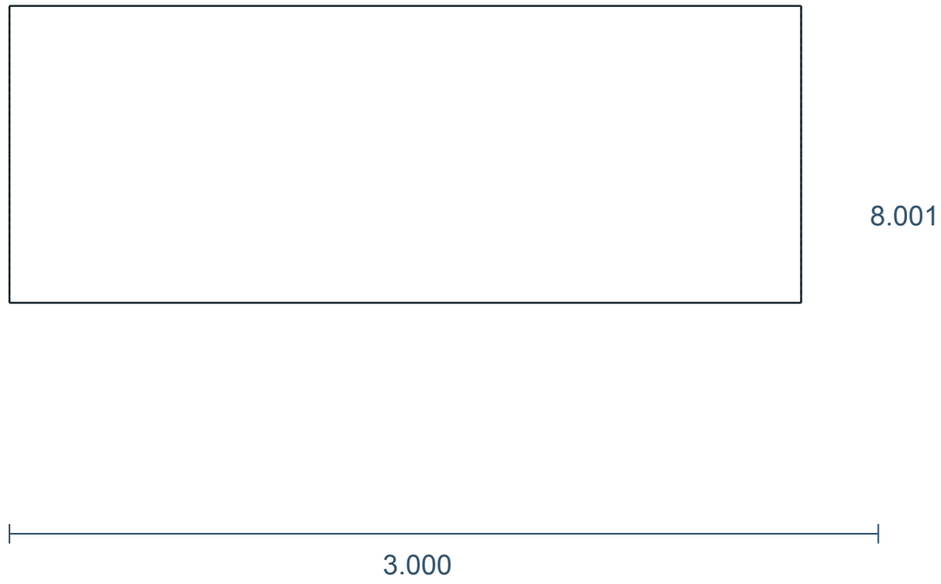
BF DOOR / D2 / lid_fastener_3

Units mm | third-angle views | prototype drawing | projection from STEP geometry

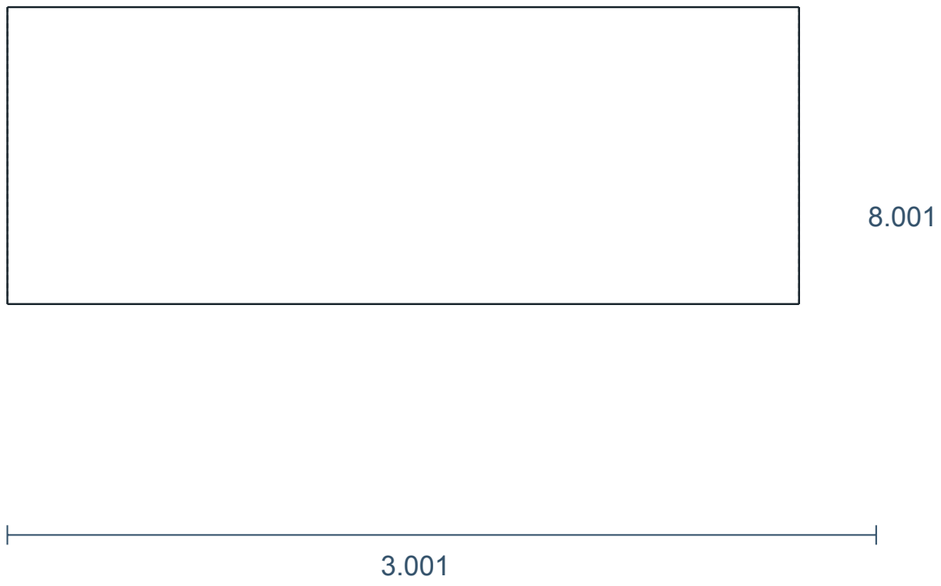
TOP



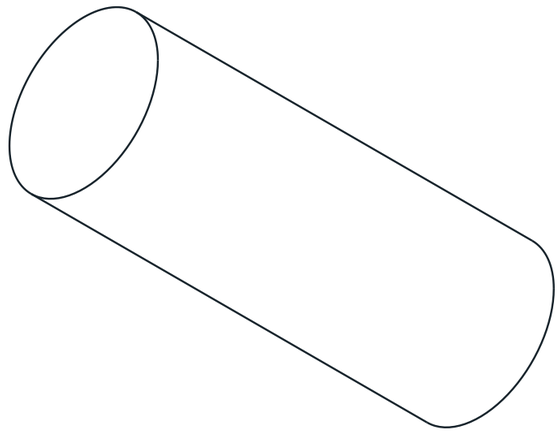
RIGHT



FRONT



ISOMETRIC



Material: M3 nylon | envelope 3.001 x 3.000 x 8.001 mm

Plastic general tolerance +/-0.25 mm; metal/ceramic tolerance requires supplier drawing.

Hole positions and boolean features: input.json. No IP, fatigue, thread-strength or fabrication-release certification.

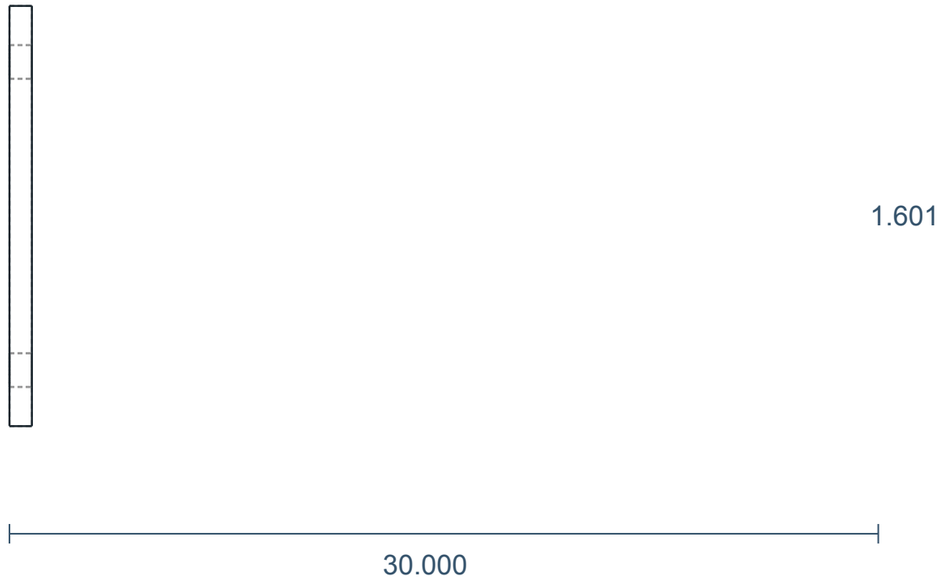
BF DOOR / D2 / main_pcb

Units mm | third-angle views | prototype drawing | projection from STEP geometry

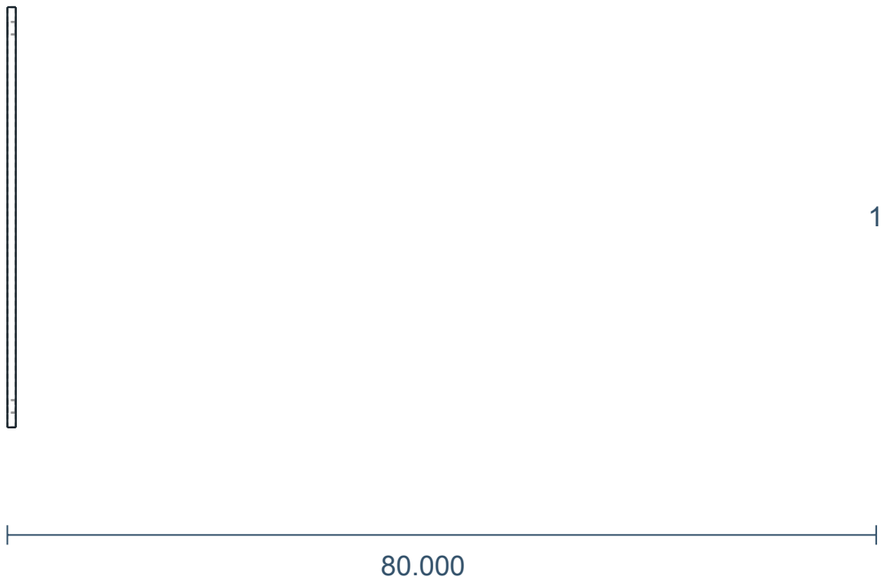
TOP



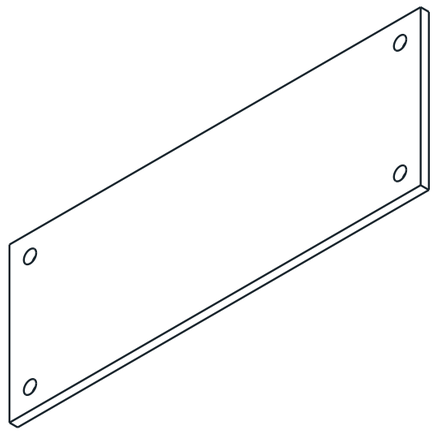
RIGHT



FRONT



ISOMETRIC



Material: FR4 1.6 mm | envelope 80.000 x 30.000 x 1.601 mm

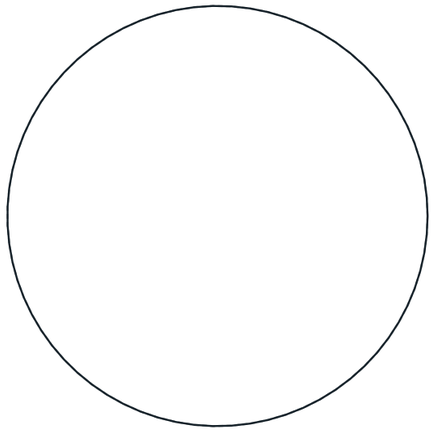
Plastic general tolerance +/-0.25 mm; metal/ceramic tolerance requires supplier drawing.

Hole positions and boolean features: input.json. No IP, fatigue, thread-strength or fabrication-release certification.

BF DOOR / D2 / pcb_screw_0

Units mm | third-angle views | prototype drawing | projection from STEP geometry

TOP



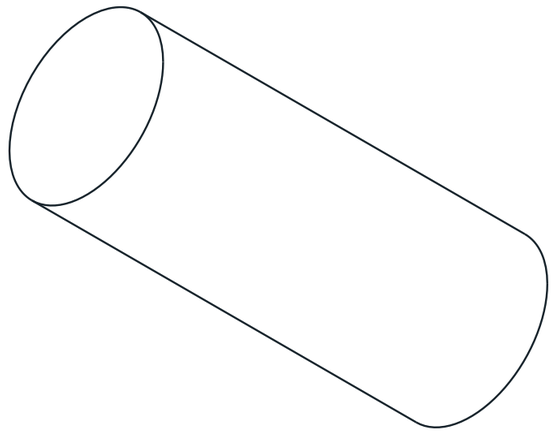
RIGHT



FRONT



ISOMETRIC



Material: M2 nylon screw envelope | envelope 2.001 x 2.000 x 5.001 mm

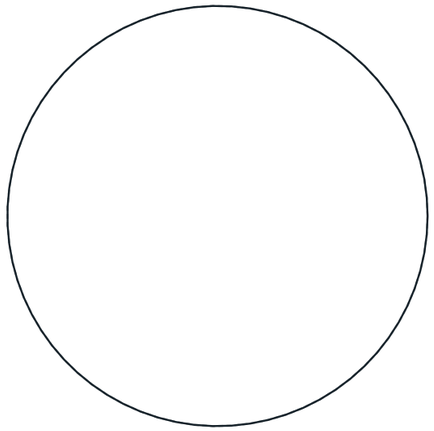
Plastic general tolerance +/-0.25 mm; metal/ceramic tolerance requires supplier drawing.

Hole positions and boolean features: input.json. No IP, fatigue, thread-strength or fabrication-release certification.

BF DOOR / D2 / pcb_screw_1

Units mm | third-angle views | prototype drawing | projection from STEP geometry

TOP



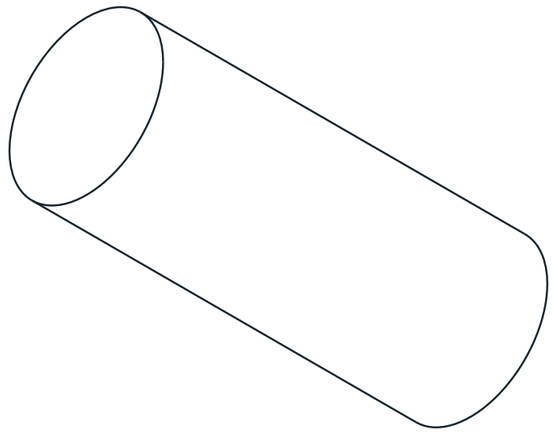
RIGHT



FRONT



ISOMETRIC



Material: M2 nylon screw envelope | envelope 2.001 x 2.000 x 5.001 mm

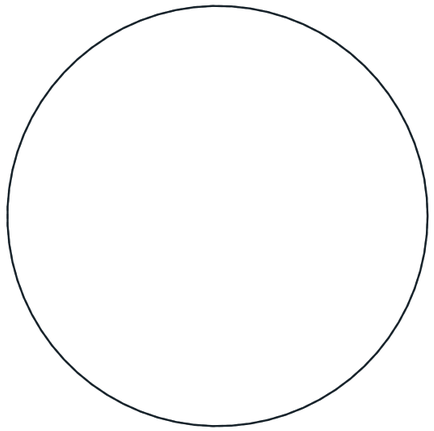
Plastic general tolerance +/-0.25 mm; metal/ceramic tolerance requires supplier drawing.

Hole positions and boolean features: input.json. No IP, fatigue, thread-strength or fabrication-release certification.

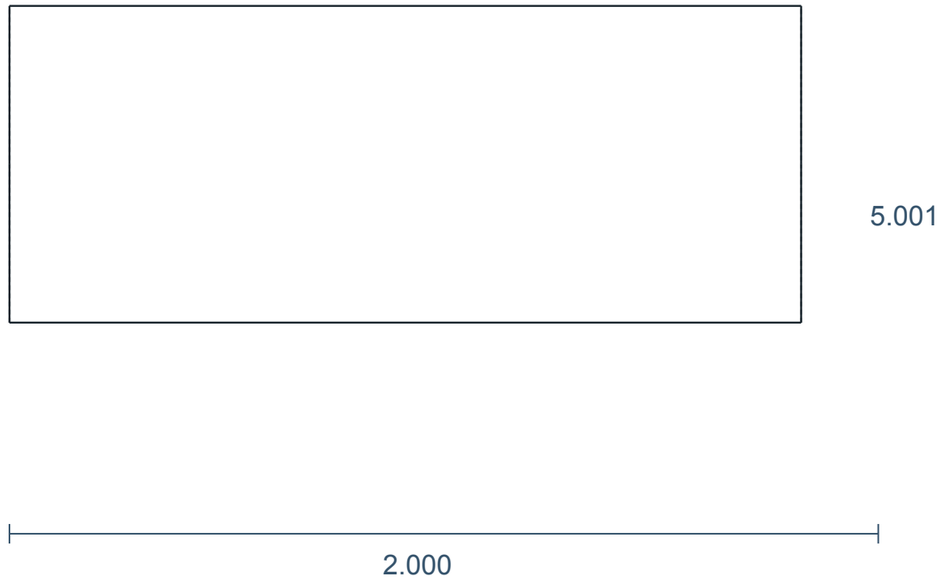
BF DOOR / D2 / pcb_screw_2

Units mm | third-angle views | prototype drawing | projection from STEP geometry

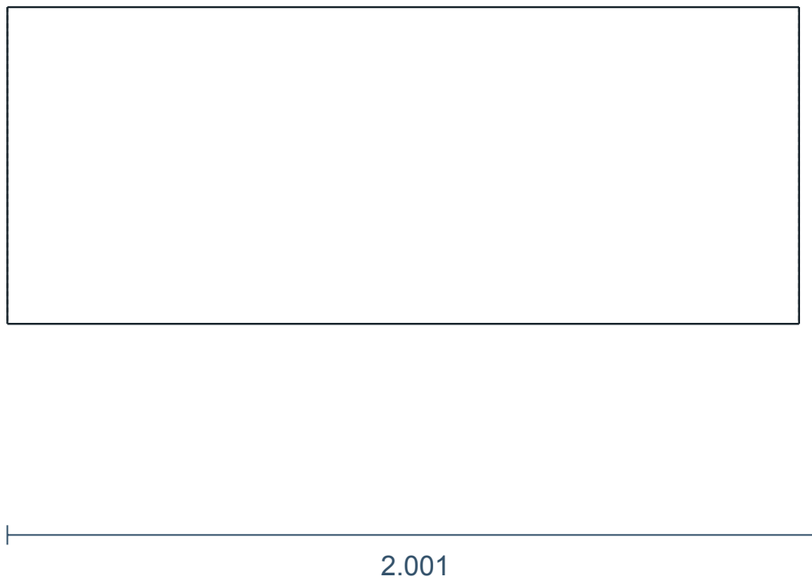
TOP



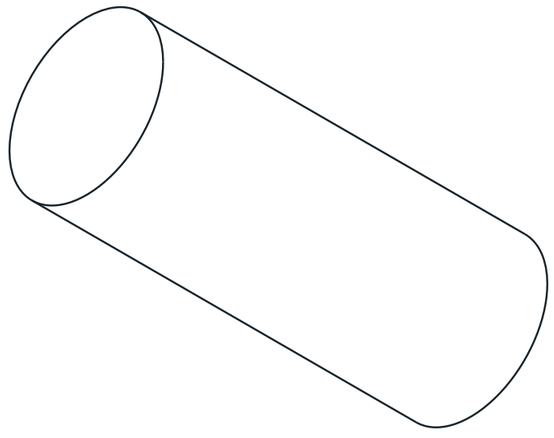
RIGHT



FRONT



ISOMETRIC



Material: M2 nylon screw envelope | envelope 2.001 x 2.000 x 5.001 mm

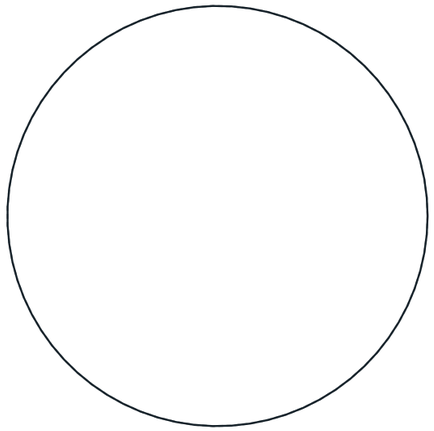
Plastic general tolerance +/-0.25 mm; metal/ceramic tolerance requires supplier drawing.

Hole positions and boolean features: input.json. No IP, fatigue, thread-strength or fabrication-release certification.

BF DOOR / D2 / pcb_screw_3

Units mm | third-angle views | prototype drawing | projection from STEP geometry

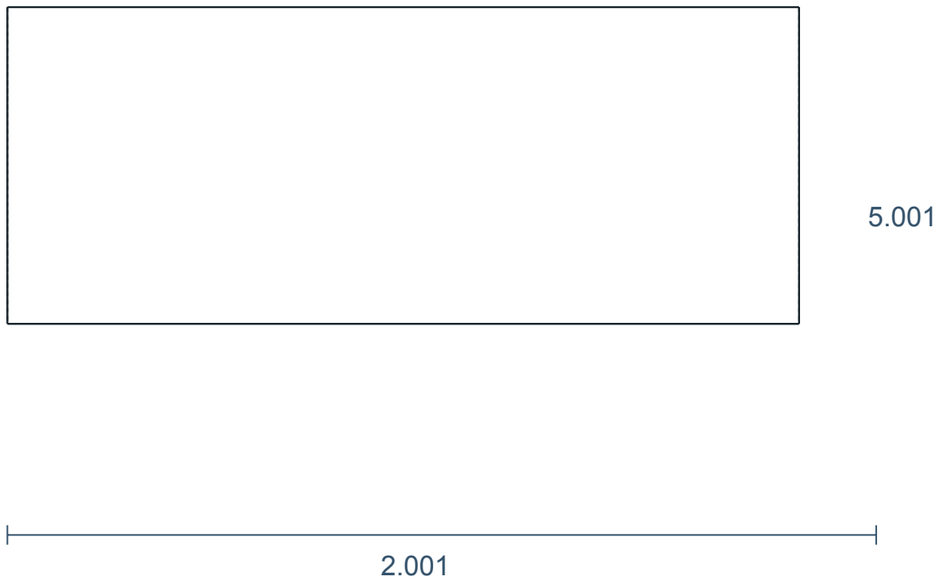
TOP



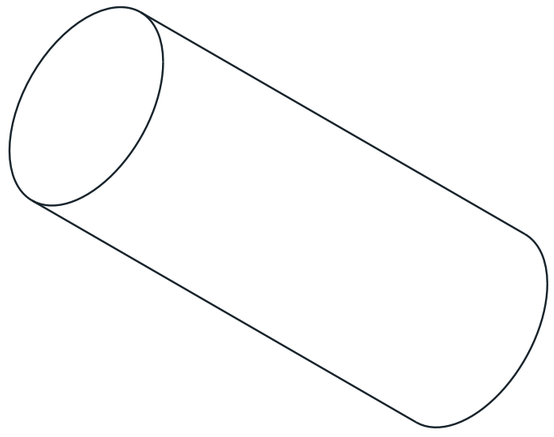
RIGHT



FRONT



ISOMETRIC



Material: M2 nylon screw envelope | envelope 2.001 x 2.000 x 5.001 mm

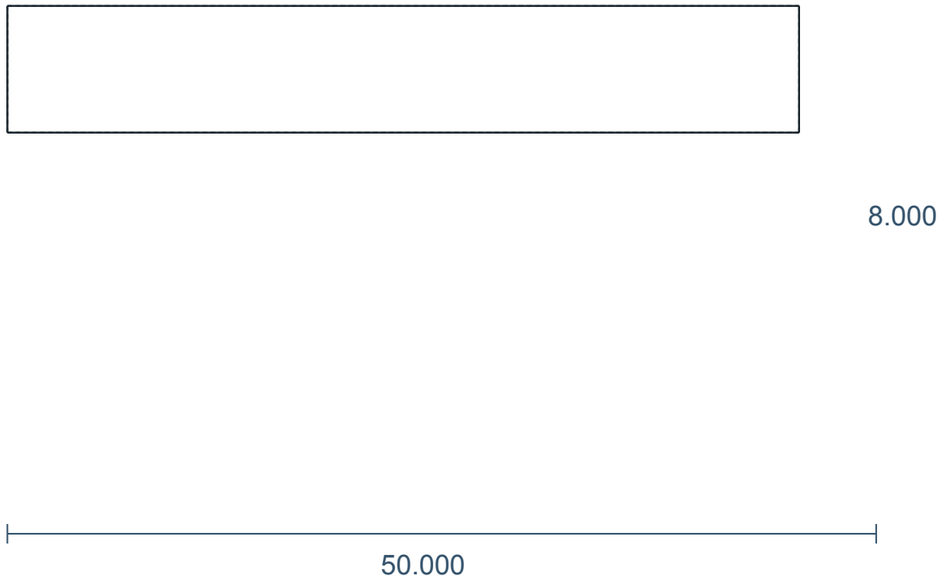
Plastic general tolerance +/-0.25 mm; metal/ceramic tolerance requires supplier drawing.

Hole positions and boolean features: input.json. No IP, fatigue, thread-strength or fabrication-release certification.

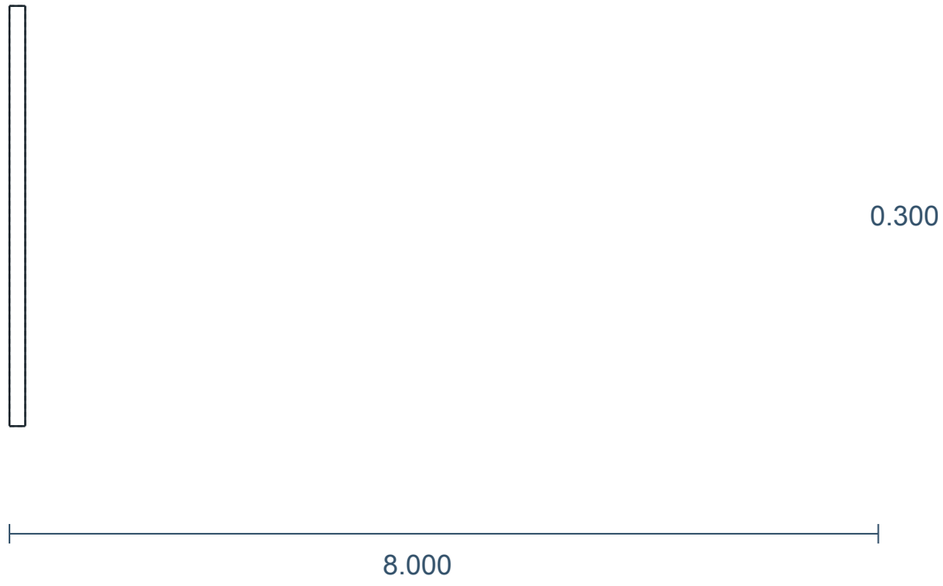
BF DOOR / D2 / pzt

Units mm | third-angle views | prototype drawing | projection from STEP geometry

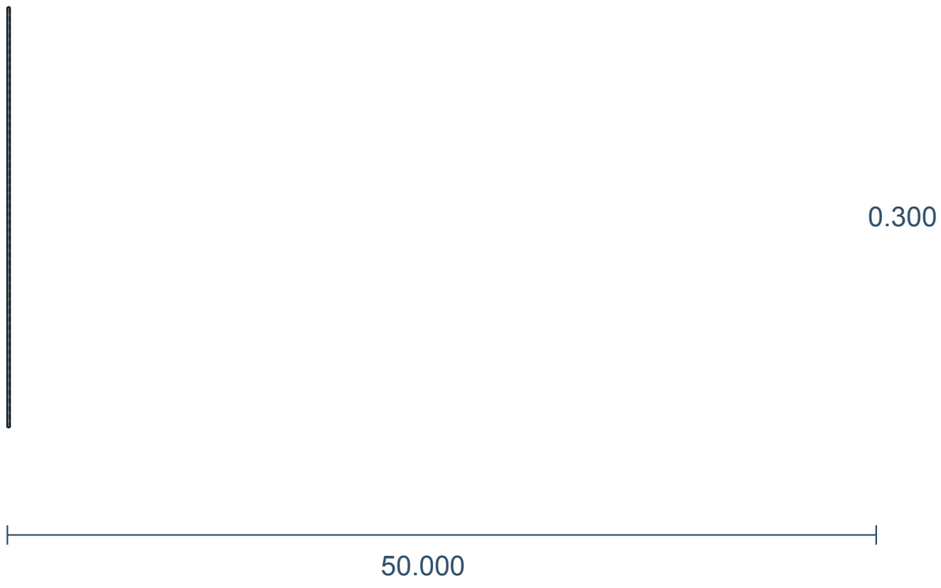
TOP



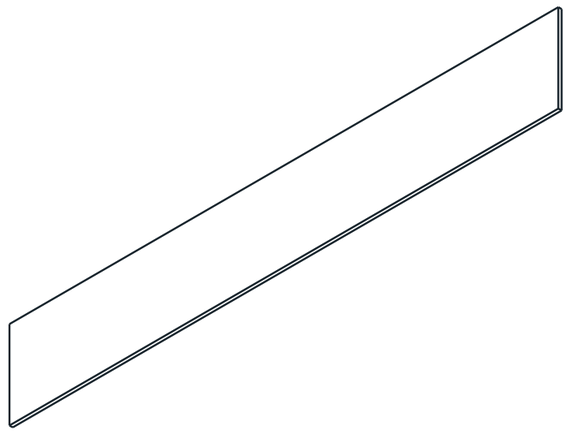
RIGHT



FRONT



ISOMETRIC



Material: PZT-5A assumed; COMSOL library | envelope 50.000 x 8.000 x 0.300 mm

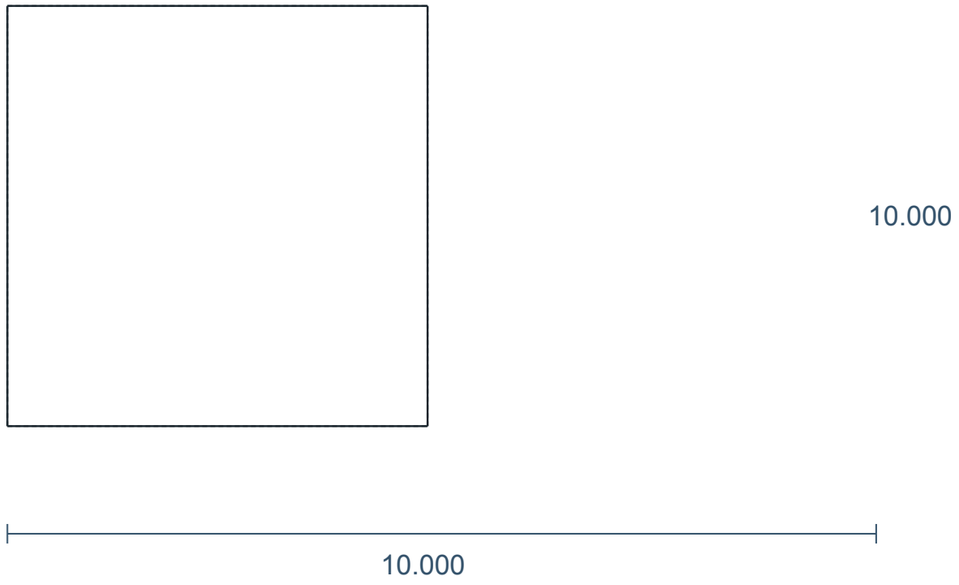
Plastic general tolerance +/-0.25 mm; metal/ceramic tolerance requires supplier drawing.

Hole positions and boolean features: input.json. No IP, fatigue, thread-strength or fabrication-release certification.

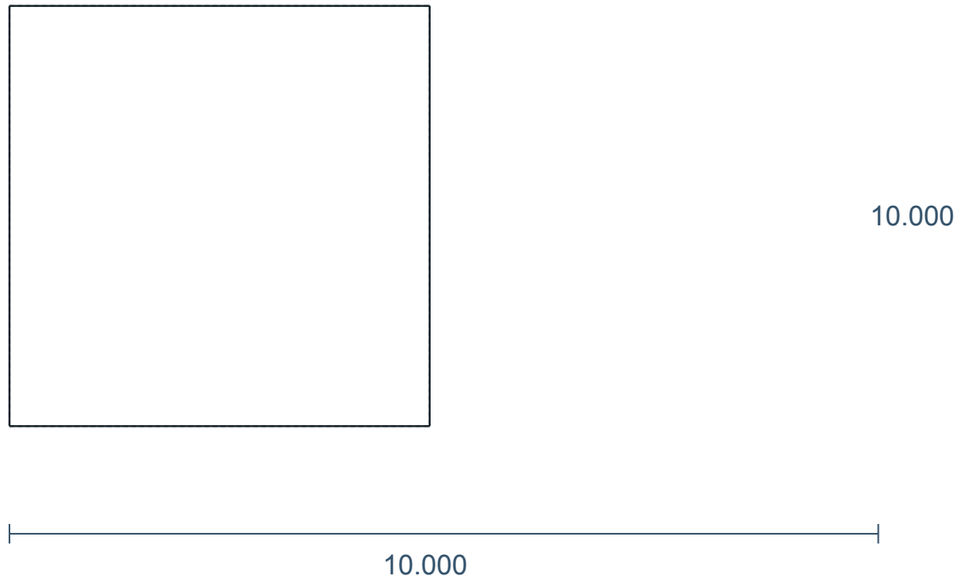
BF DOOR / D2 / tip_magnet

Units mm | third-angle views | prototype drawing | projection from STEP geometry

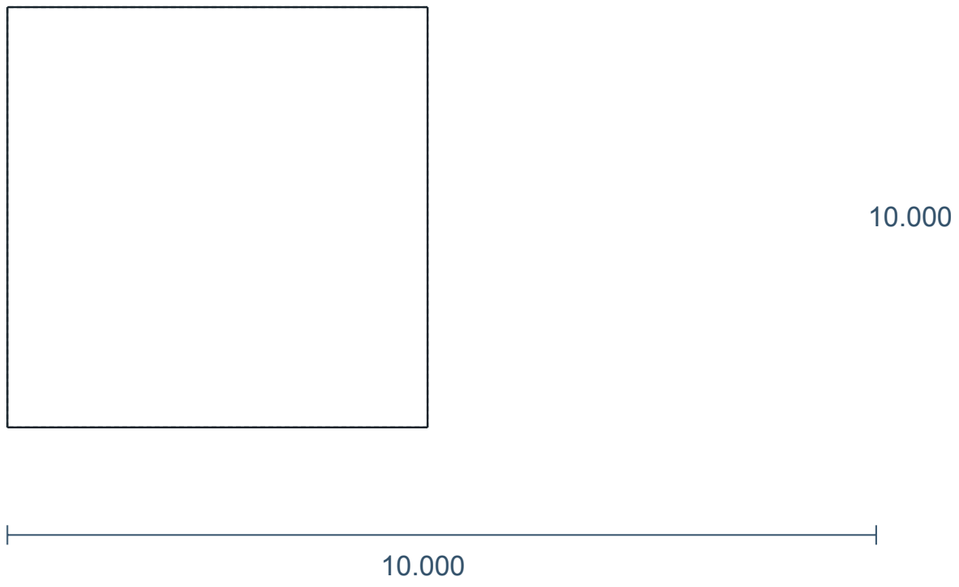
TOP



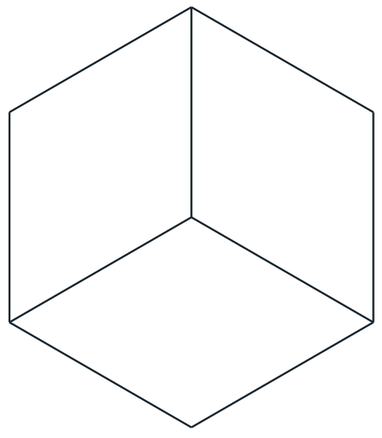
RIGHT



FRONT



ISOMETRIC



Material: NdFeB N42 nominal Br=1.25 T | envelope 10.000 x 10.000 x 10.000 mm

Plastic general tolerance +/-0.25 mm; metal/ceramic tolerance requires supplier drawing.

Hole positions and boolean features: input.json. No IP, fatigue, thread-strength or fabrication-release certification.